

Abstract Linear Algebra

Prof. Lanier

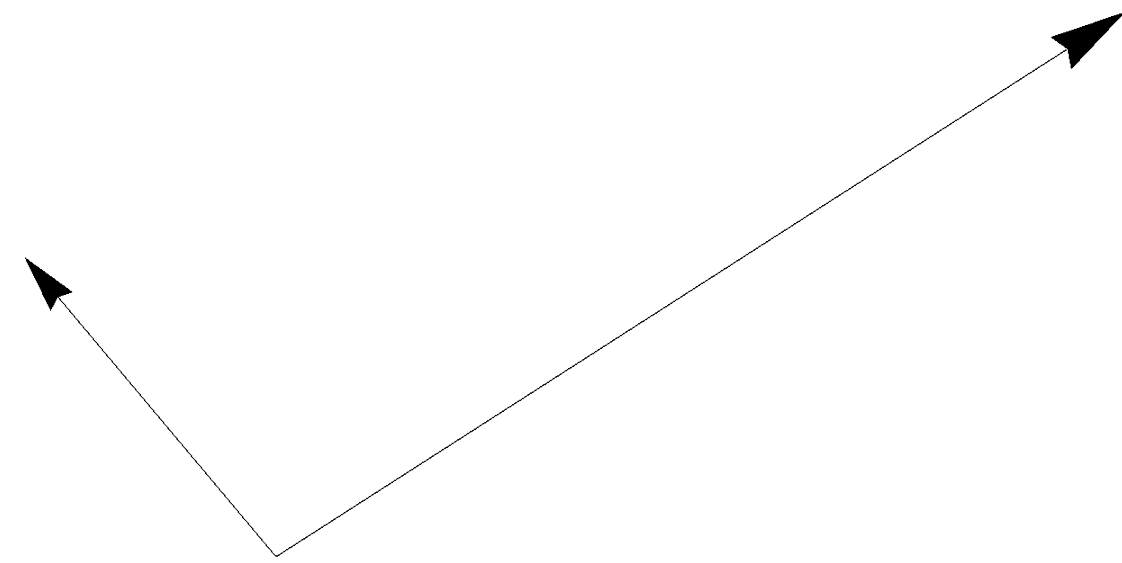
jlancier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

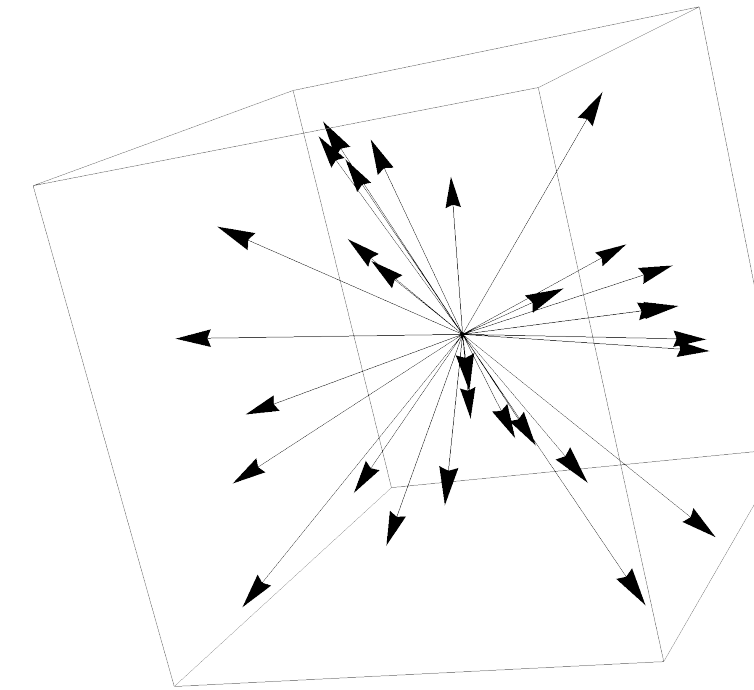
Next up:

- Today and Friday: Chapter 3
- First problem swap problems due *tonight*.
- Quizzes 4 & 5 on Friday (solving systems to vector form, finding inverses)
- Problem session Friday, 3-5pm, Ryerson 177.
- Homework 3 is due Friday / Saturday.
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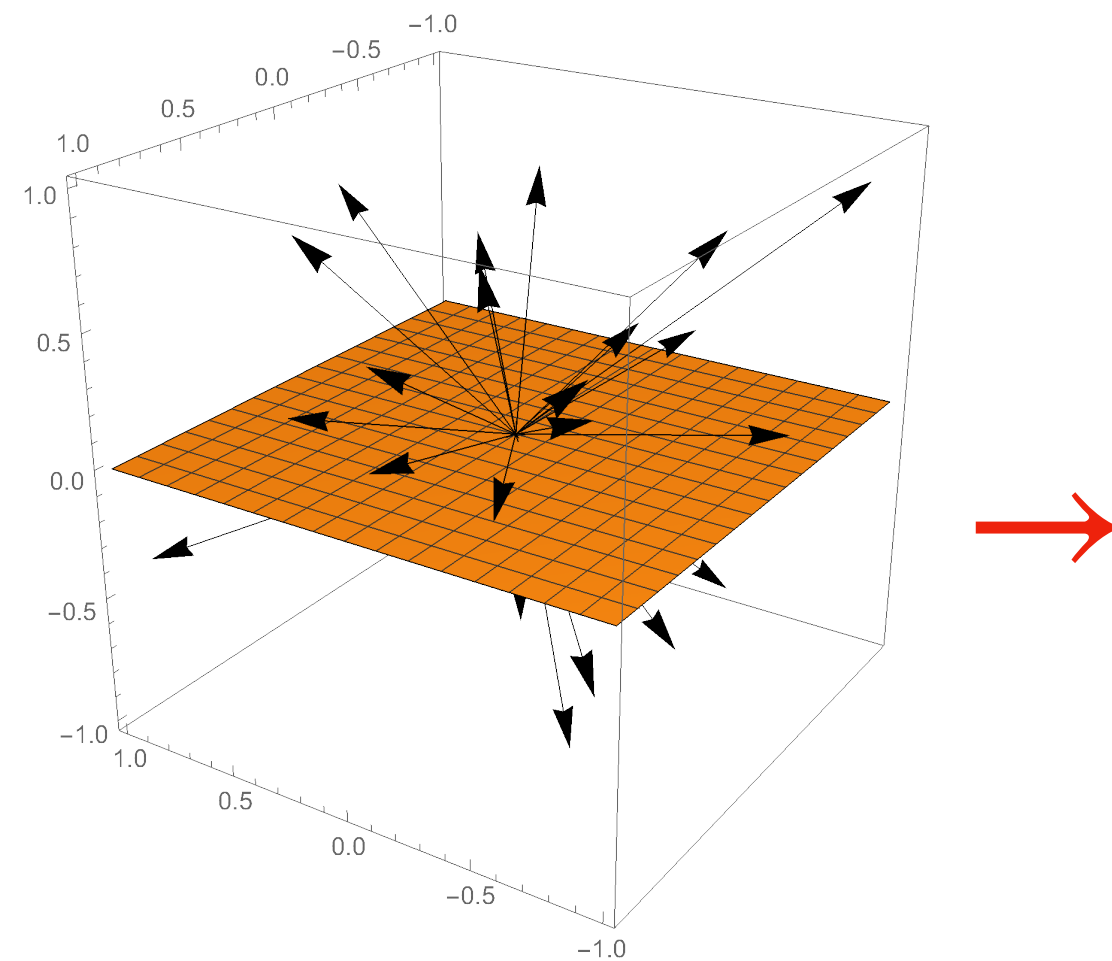
What is Linear Algebra?



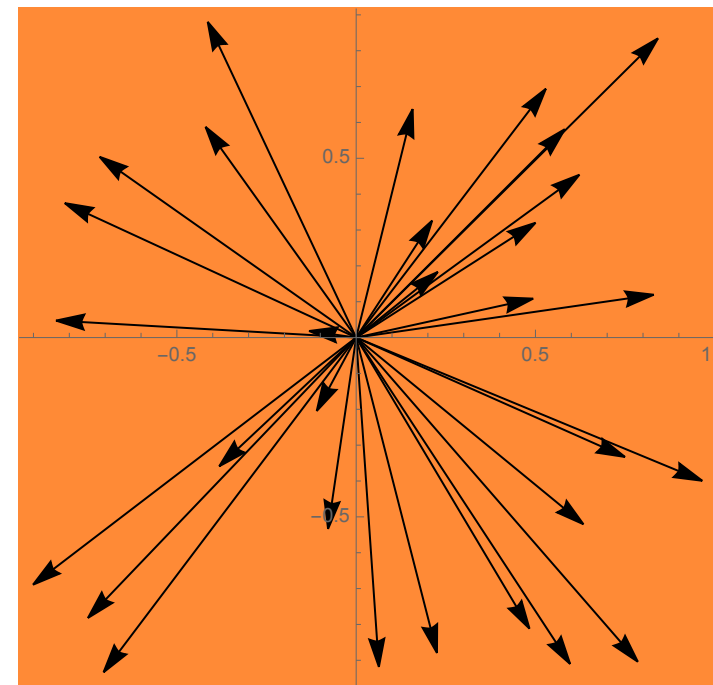
vectors



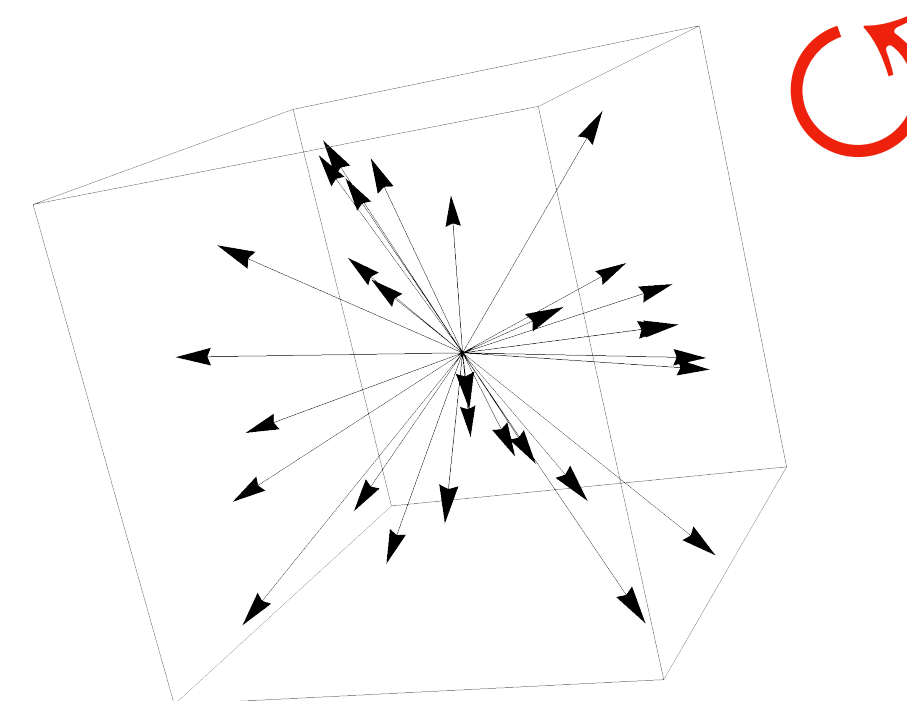
vector spaces



linear maps between
vector spaces

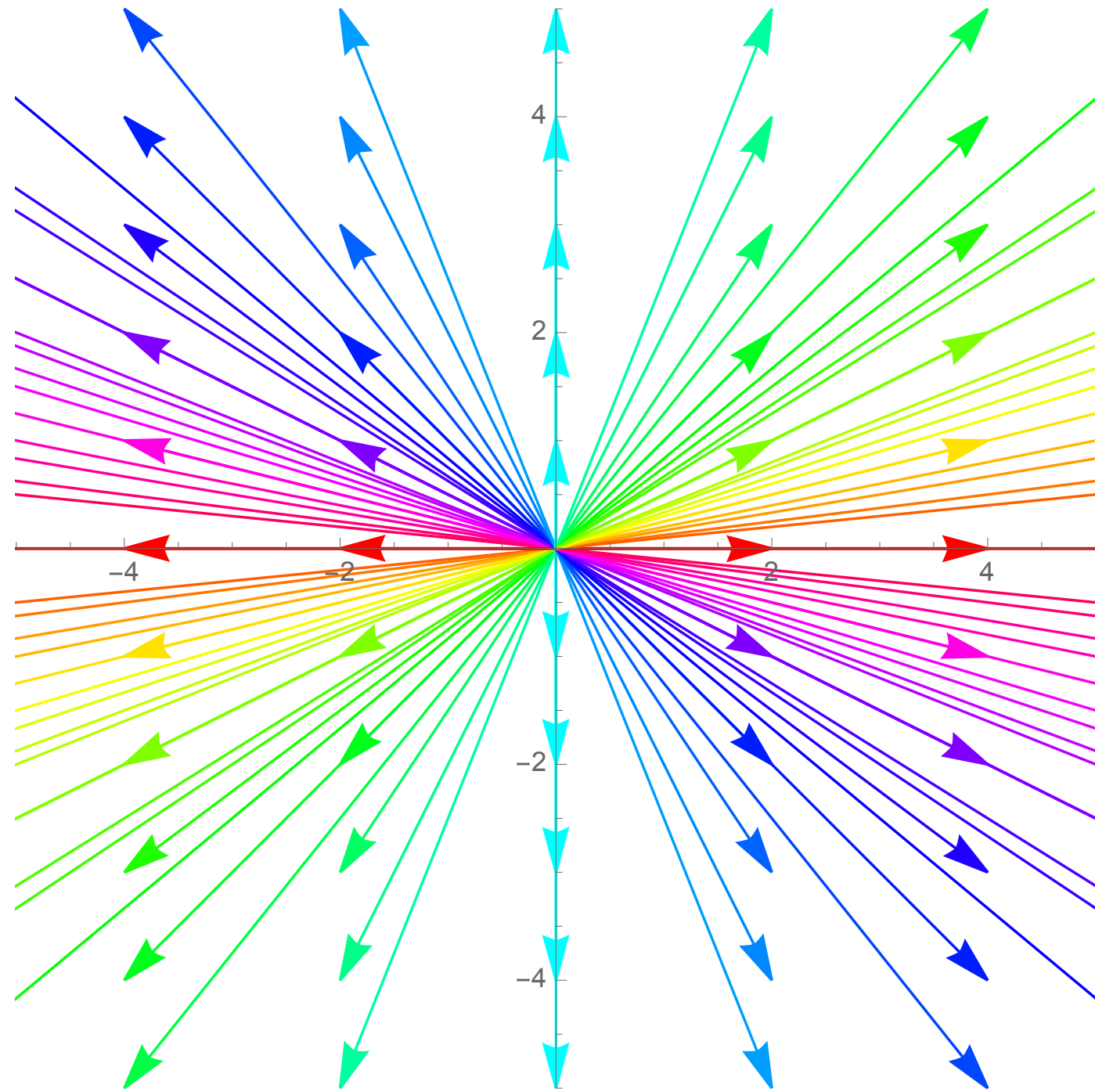


linear self-maps of
vector spaces



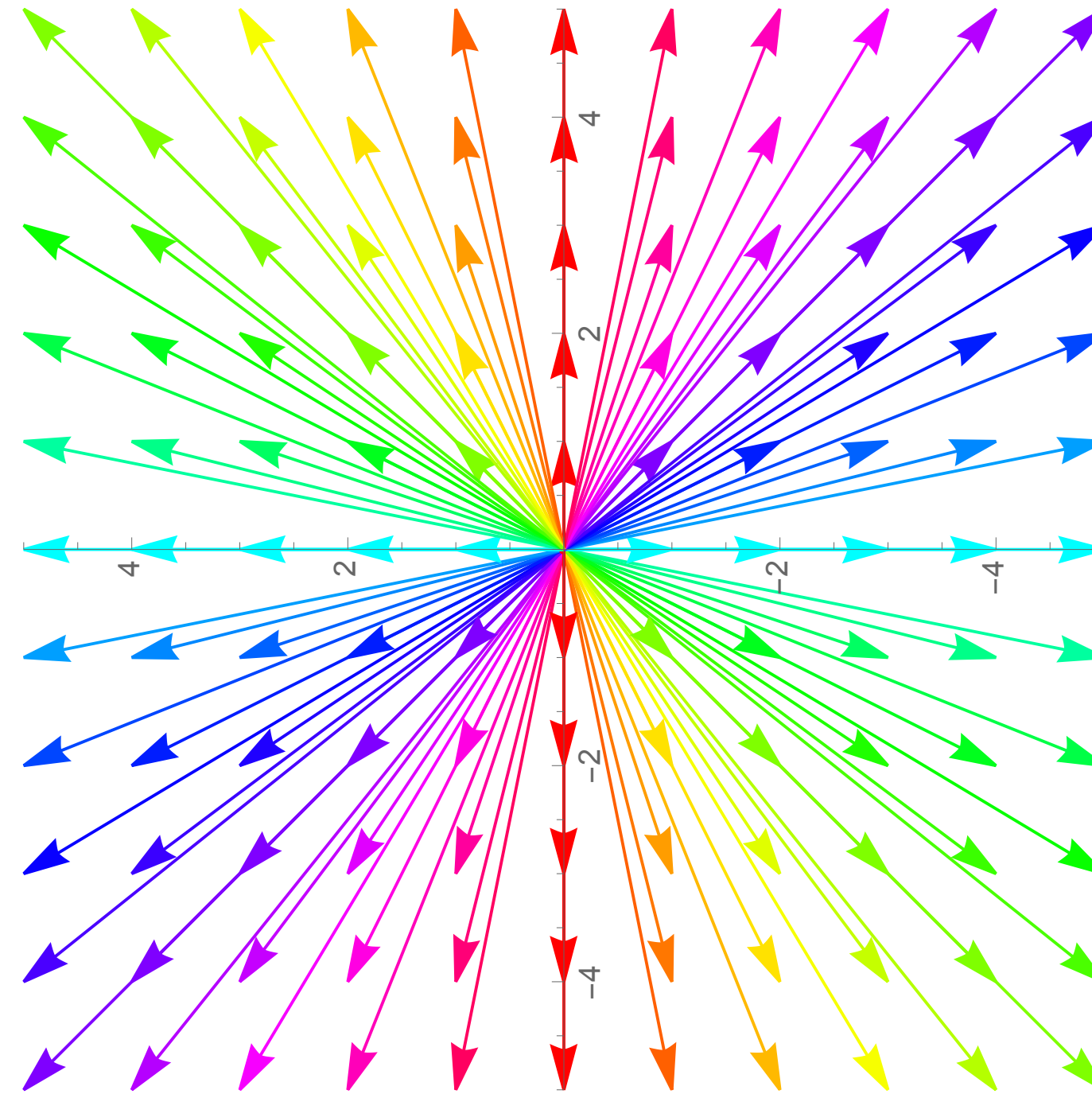
Ch 3

determinant: “global stretch factor” of a linear transformation
 $\det(T)$ is the factor by which T scales signed area (or n -dimensional volume).



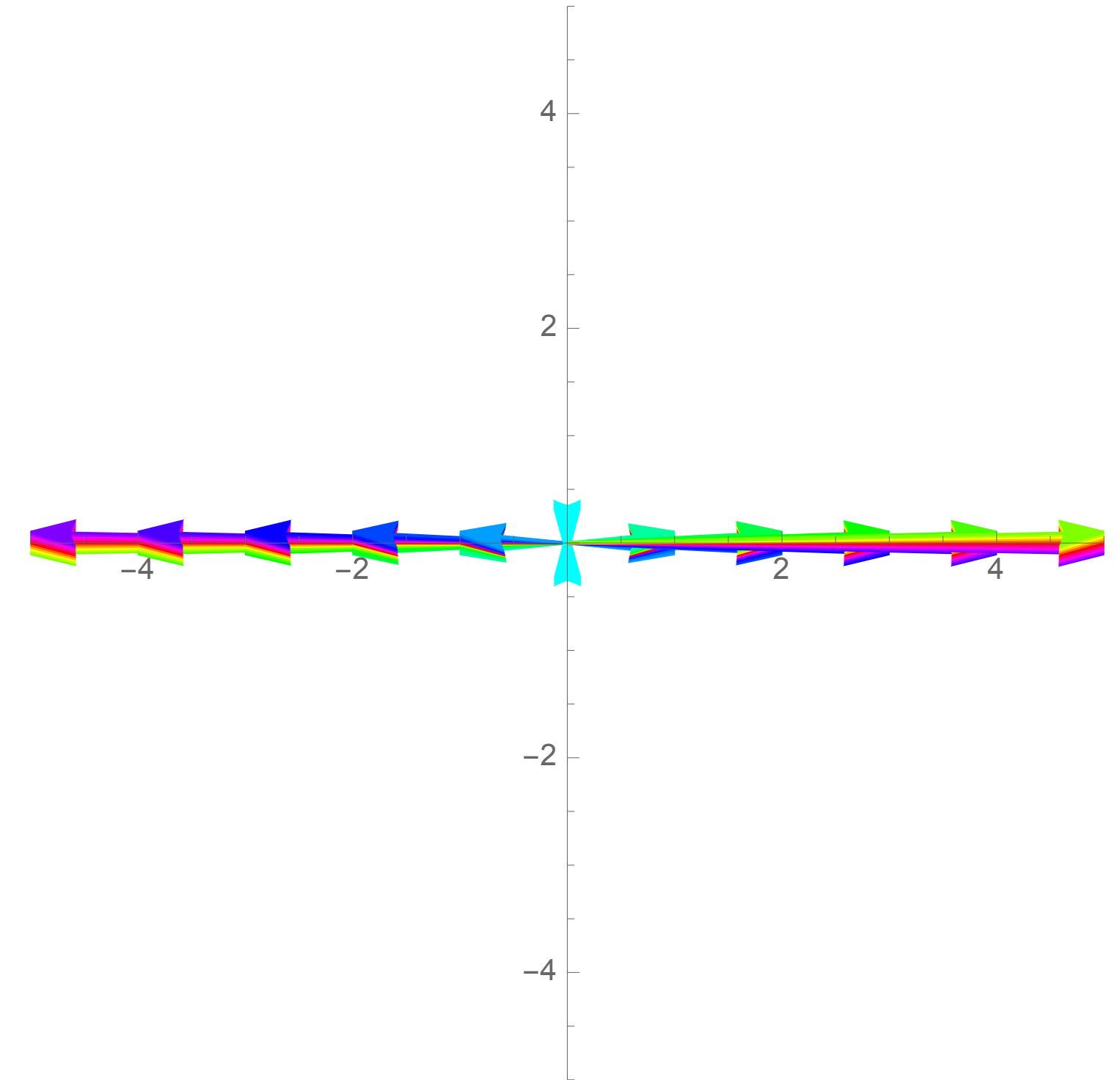
$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal



$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

rotation



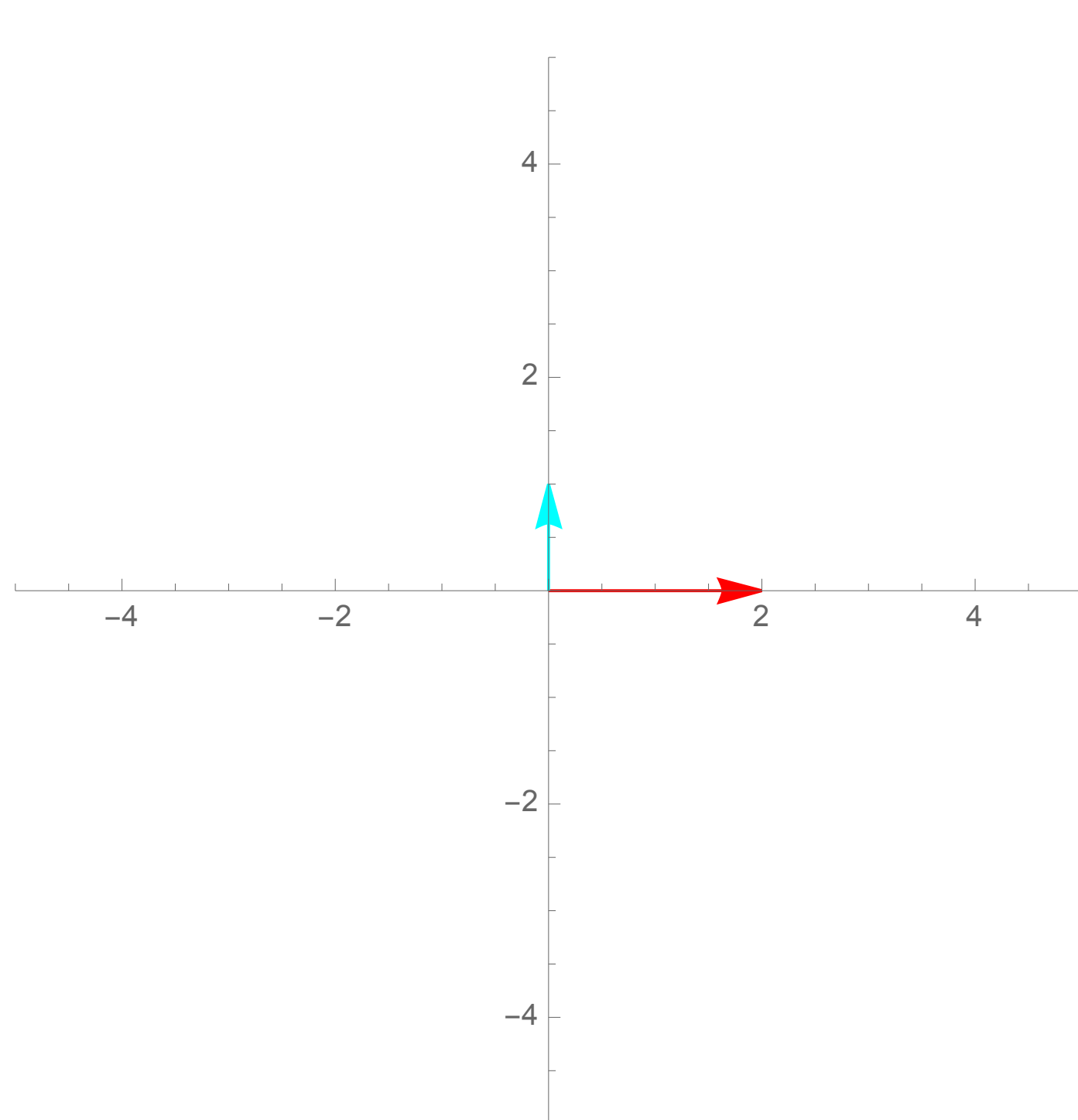
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projection

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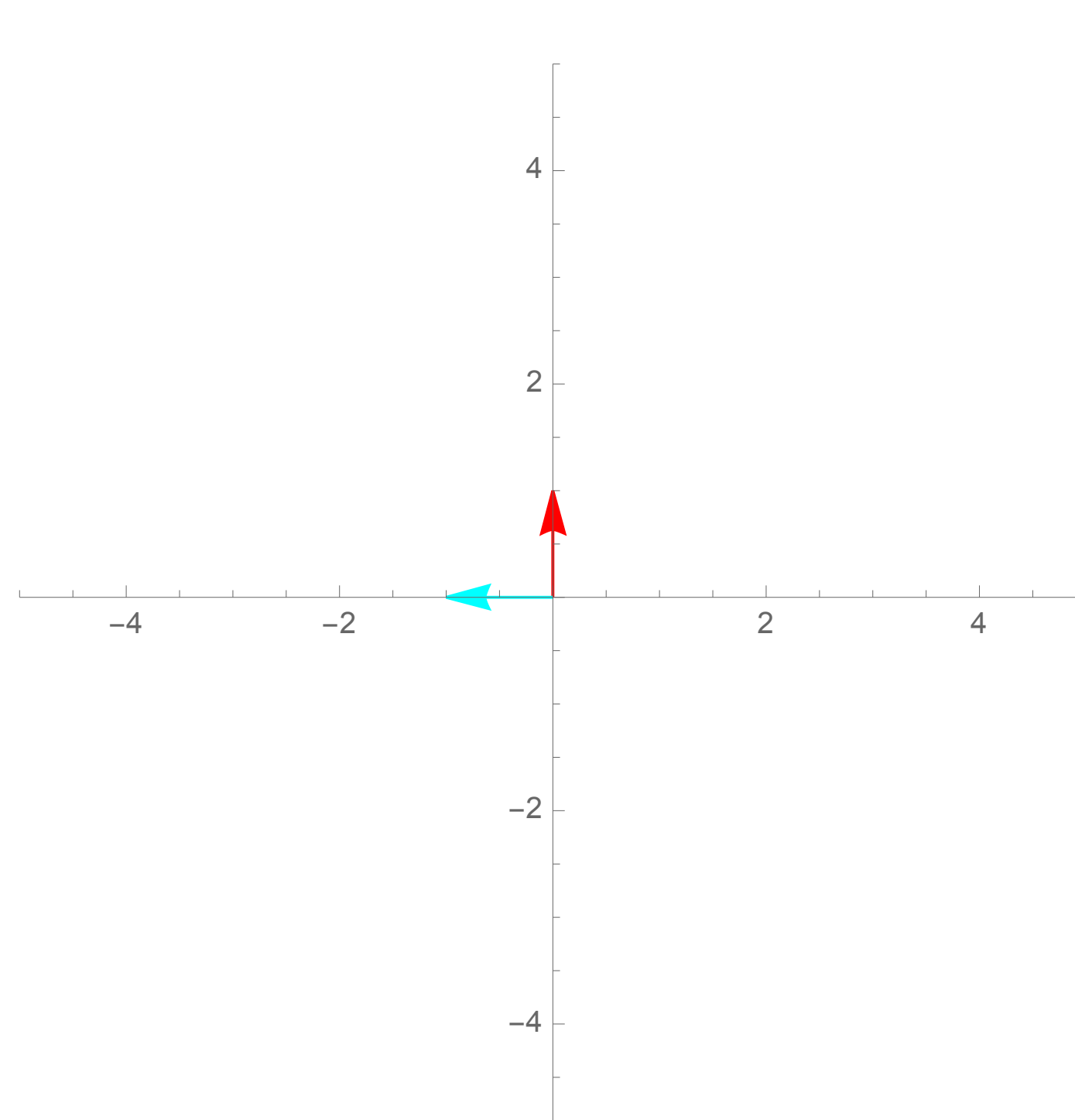
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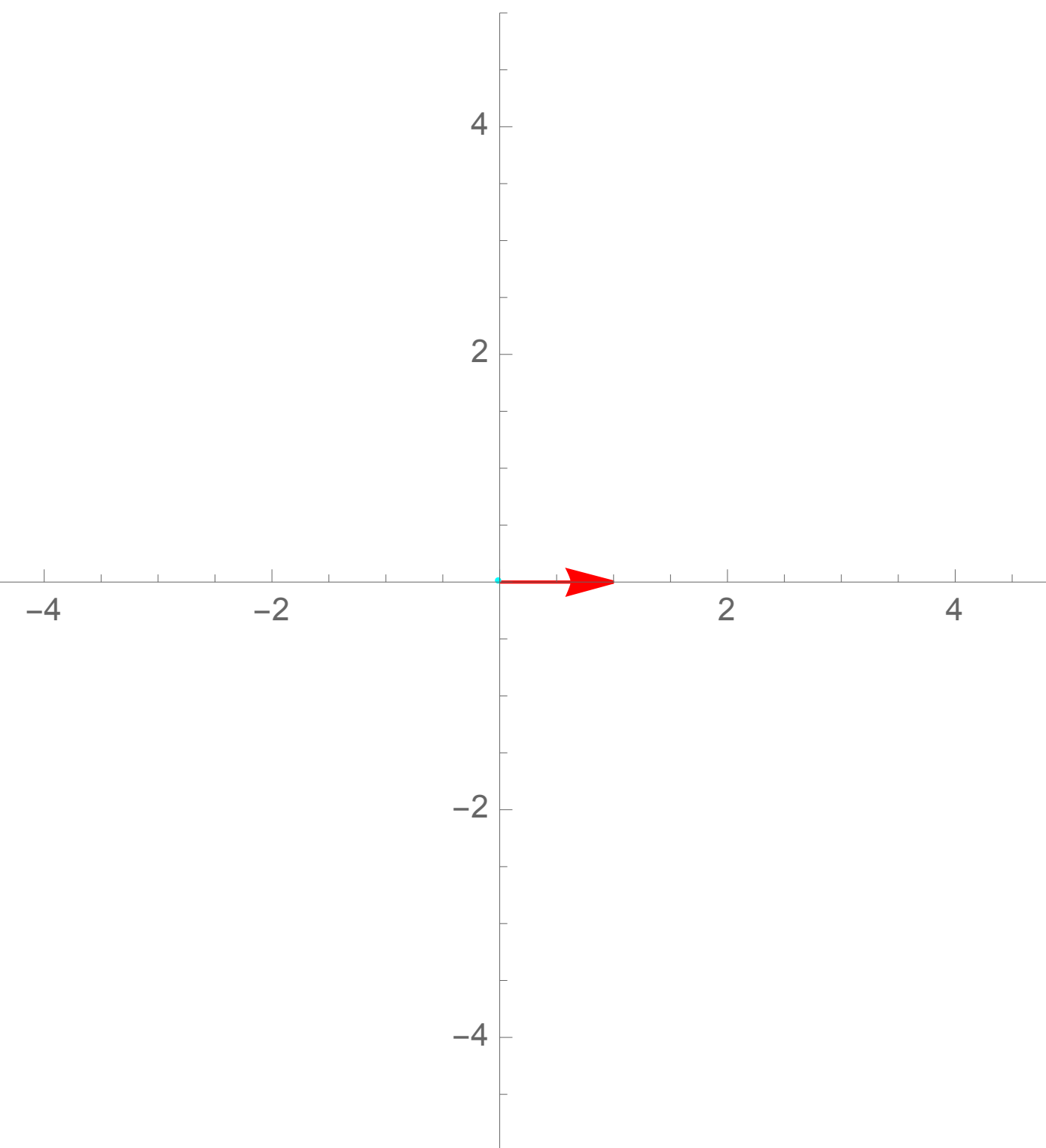
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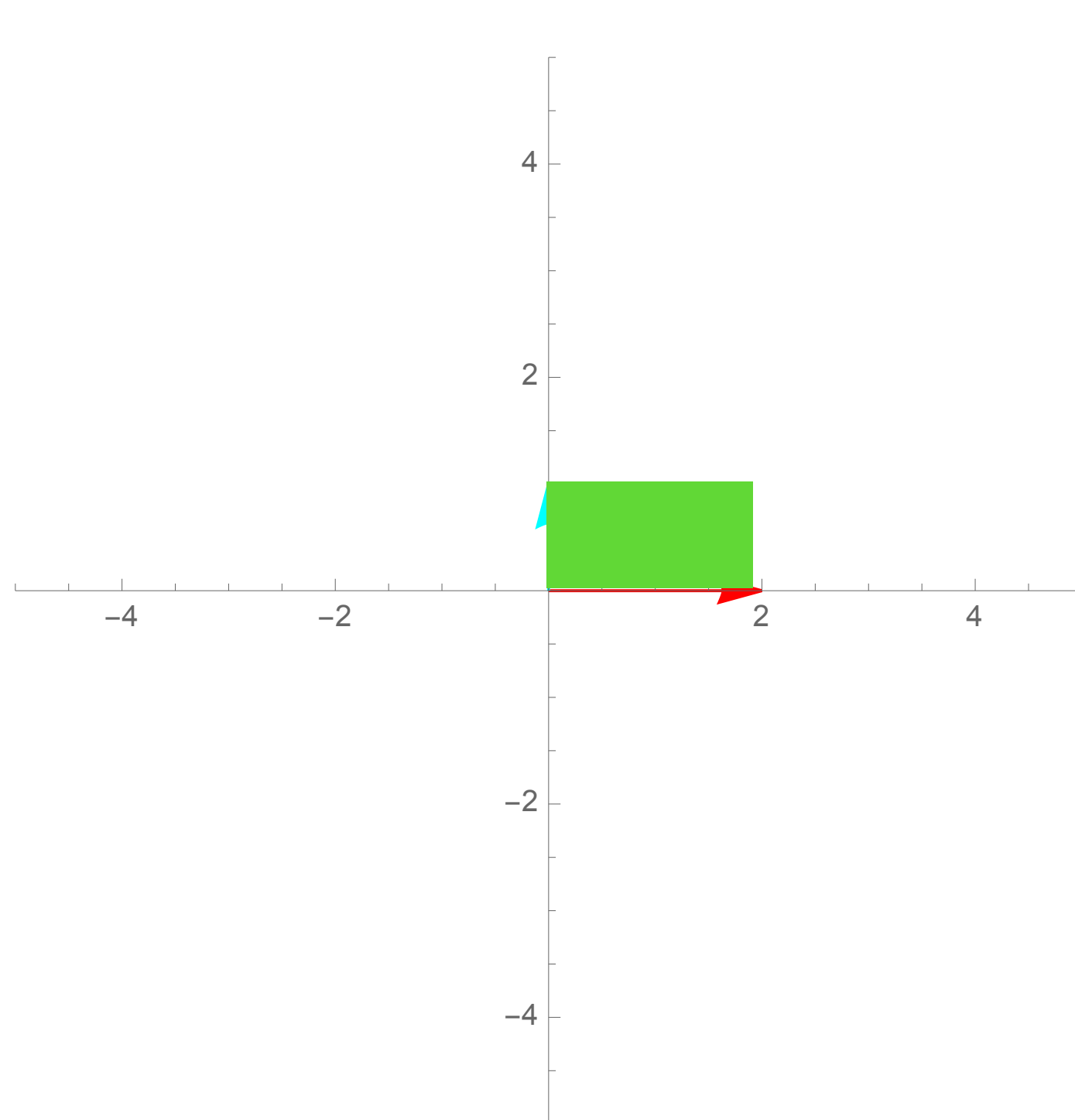
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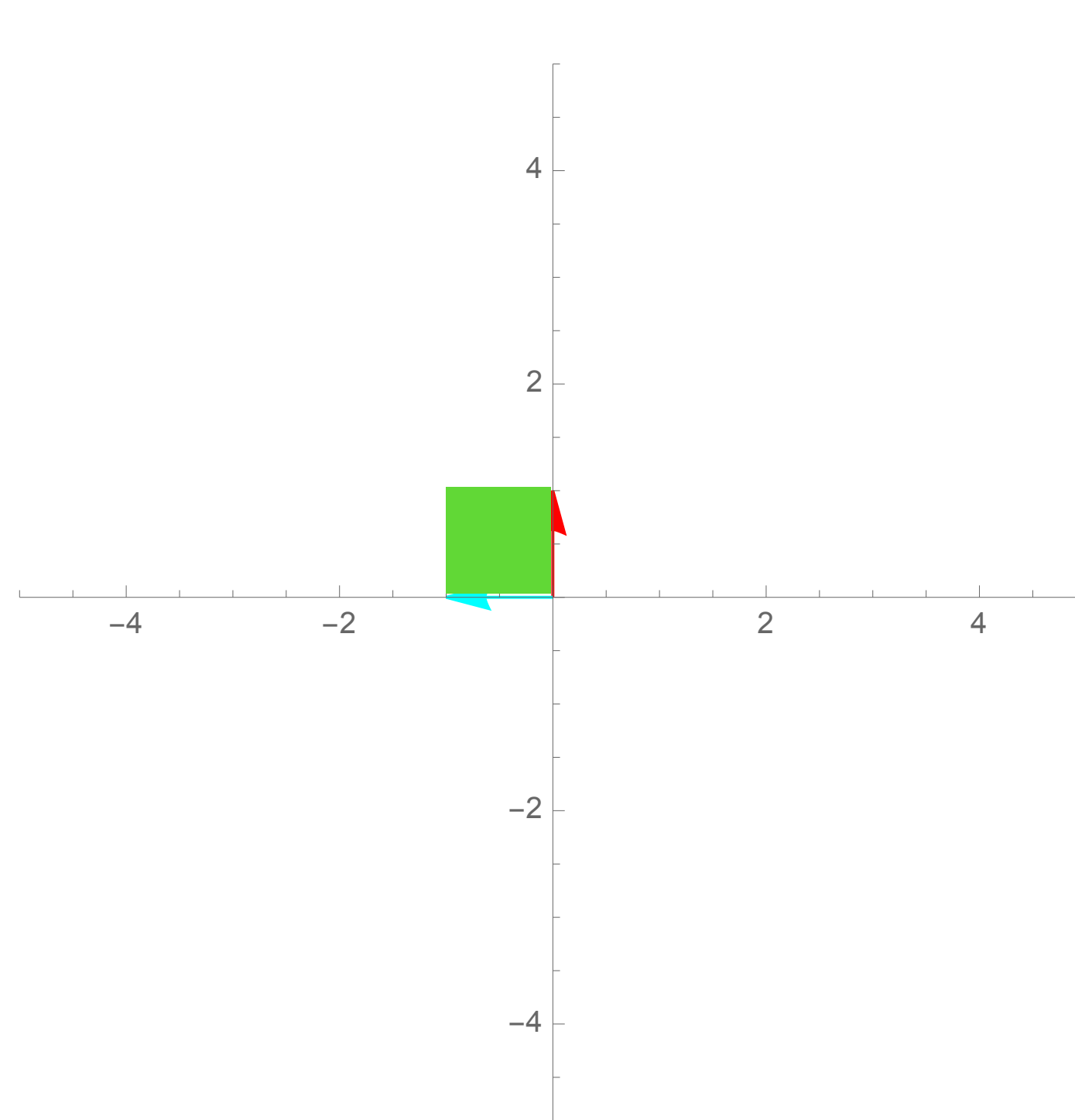
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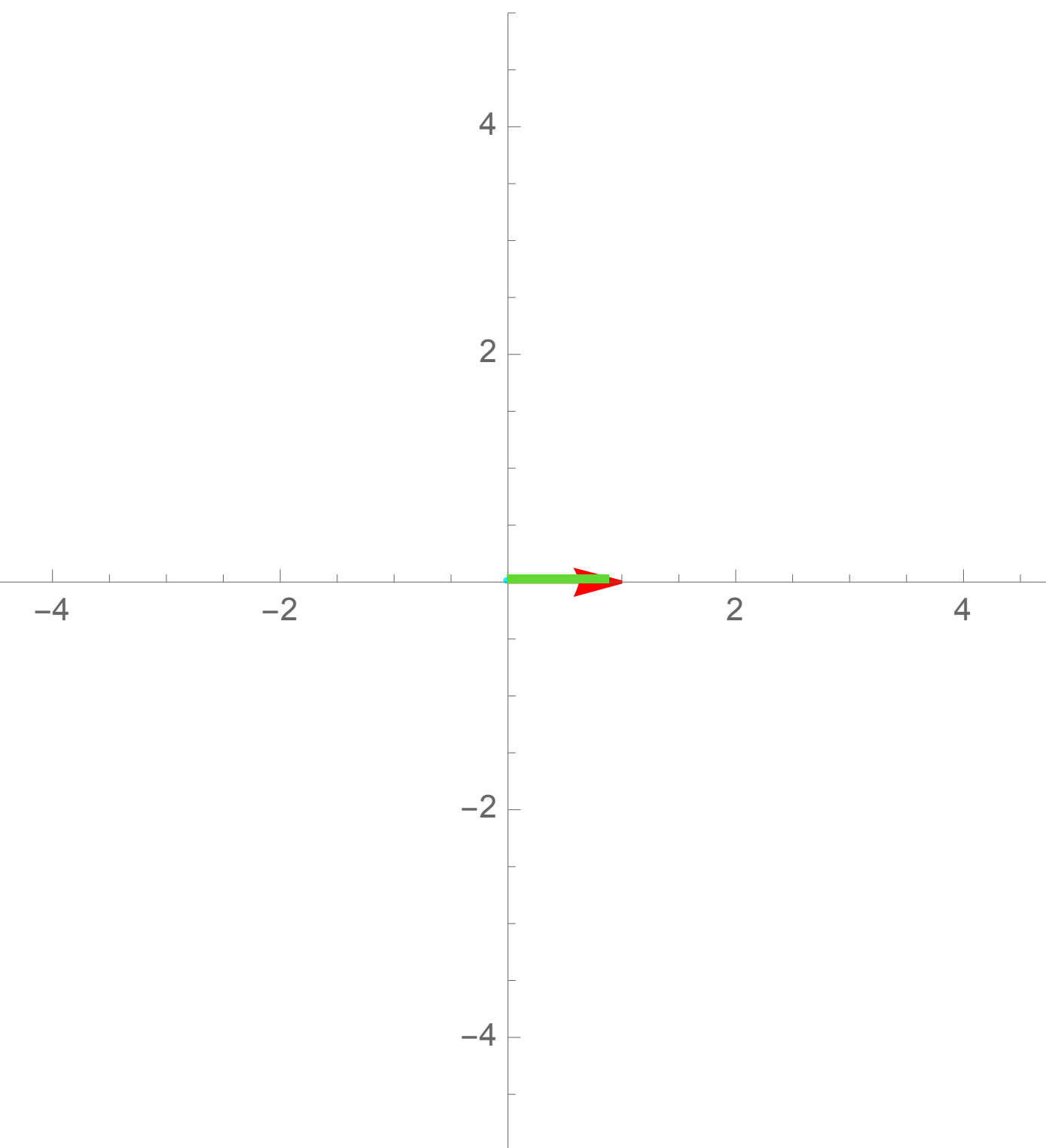
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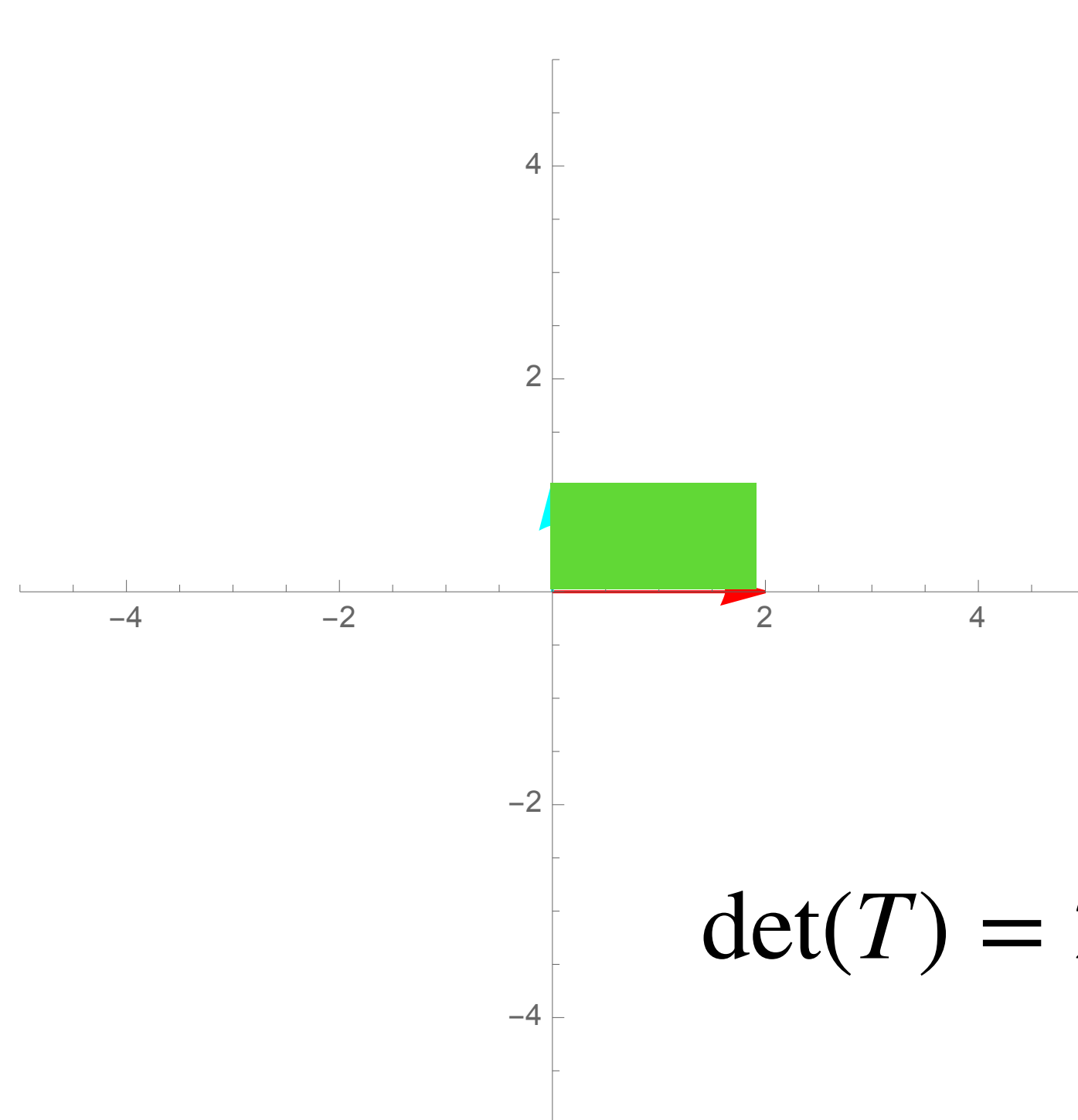
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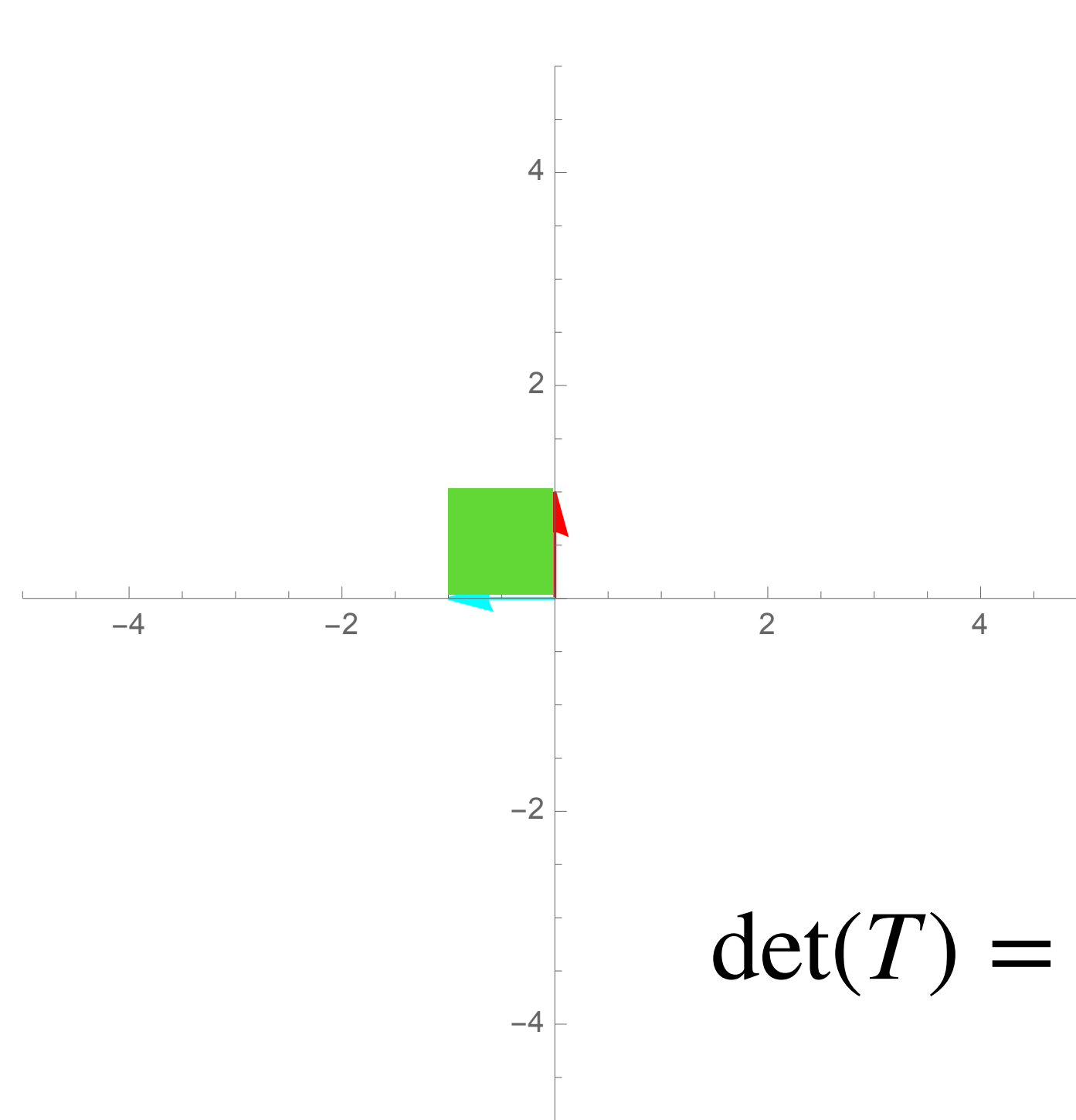
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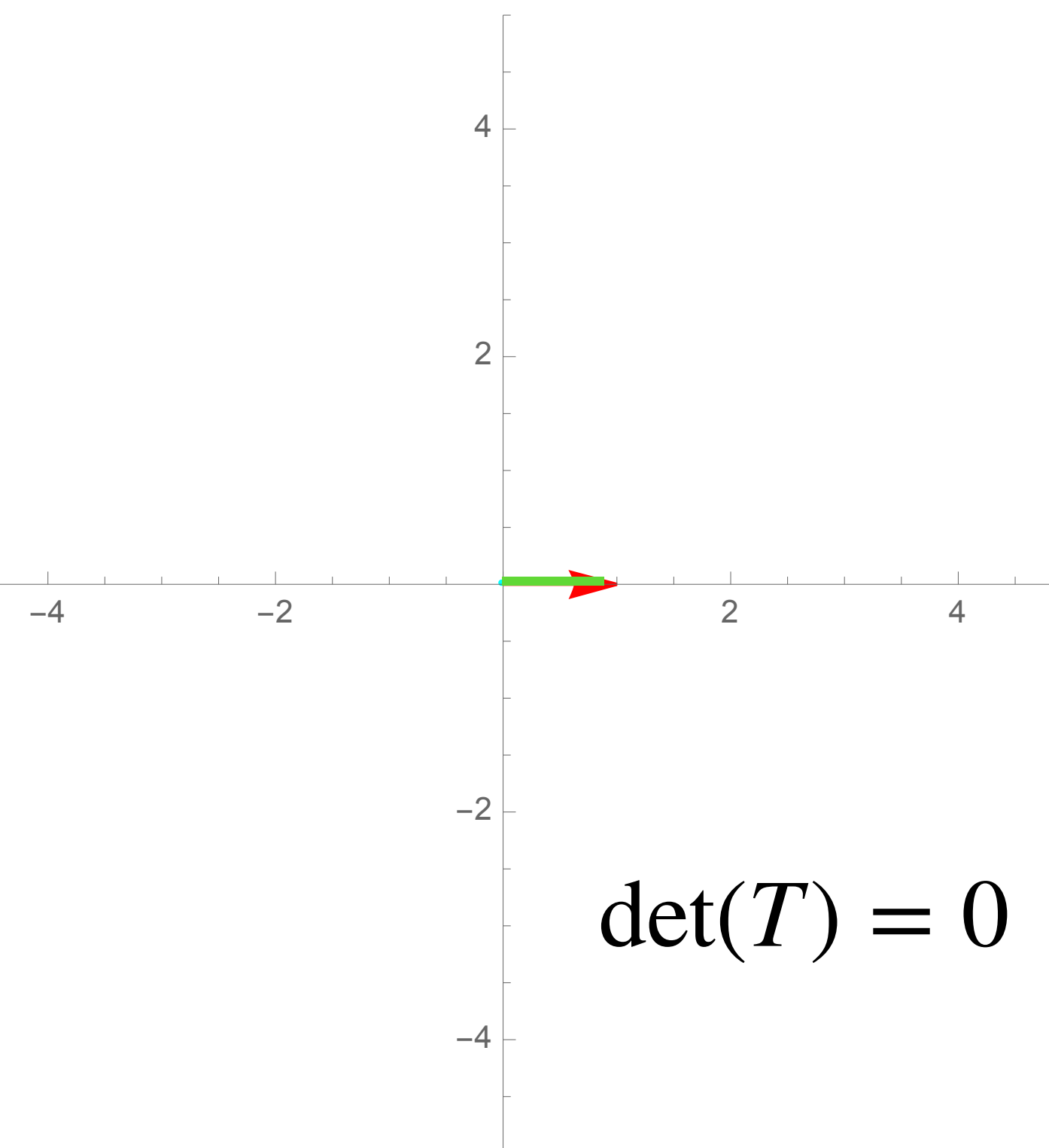
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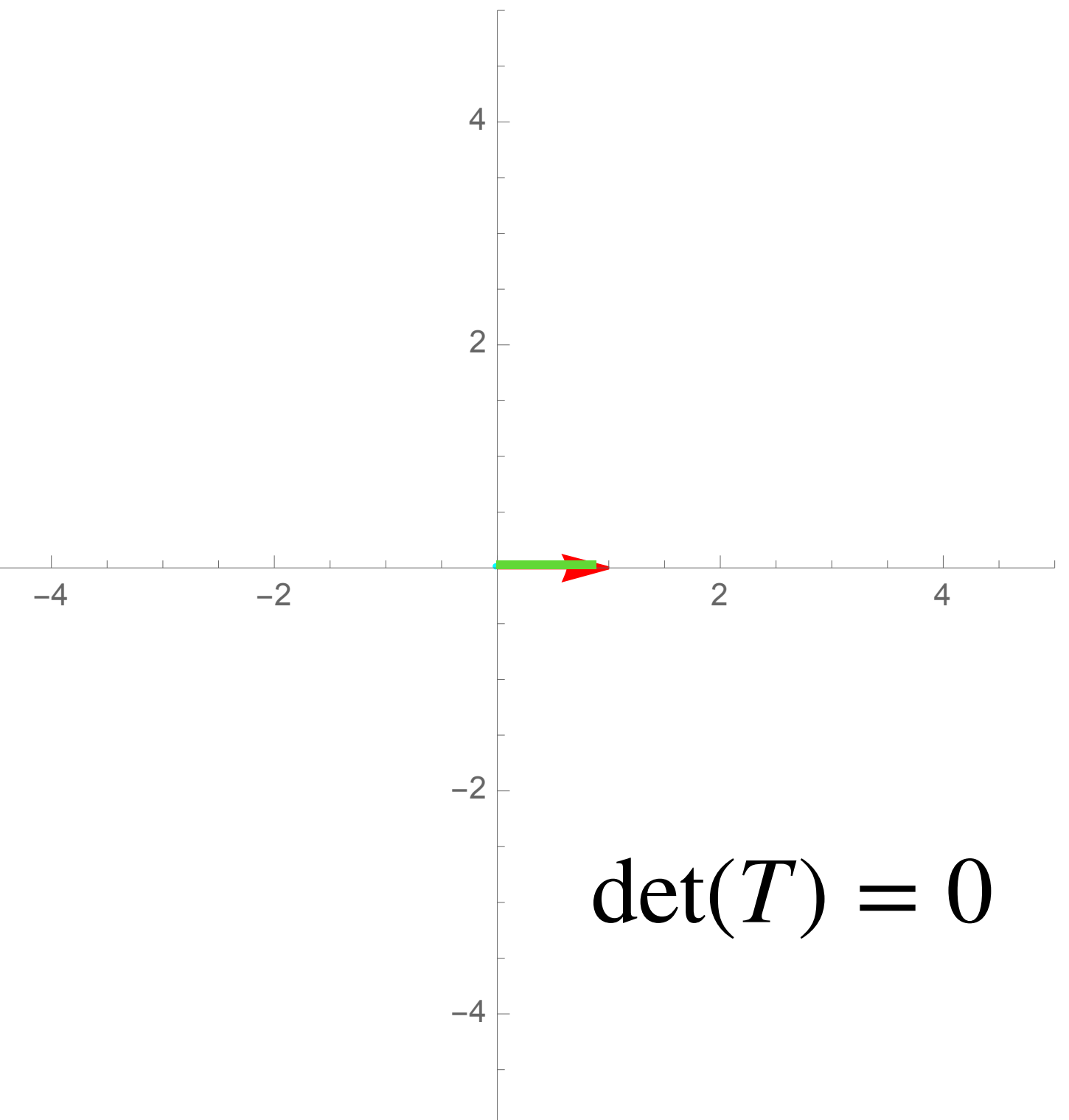
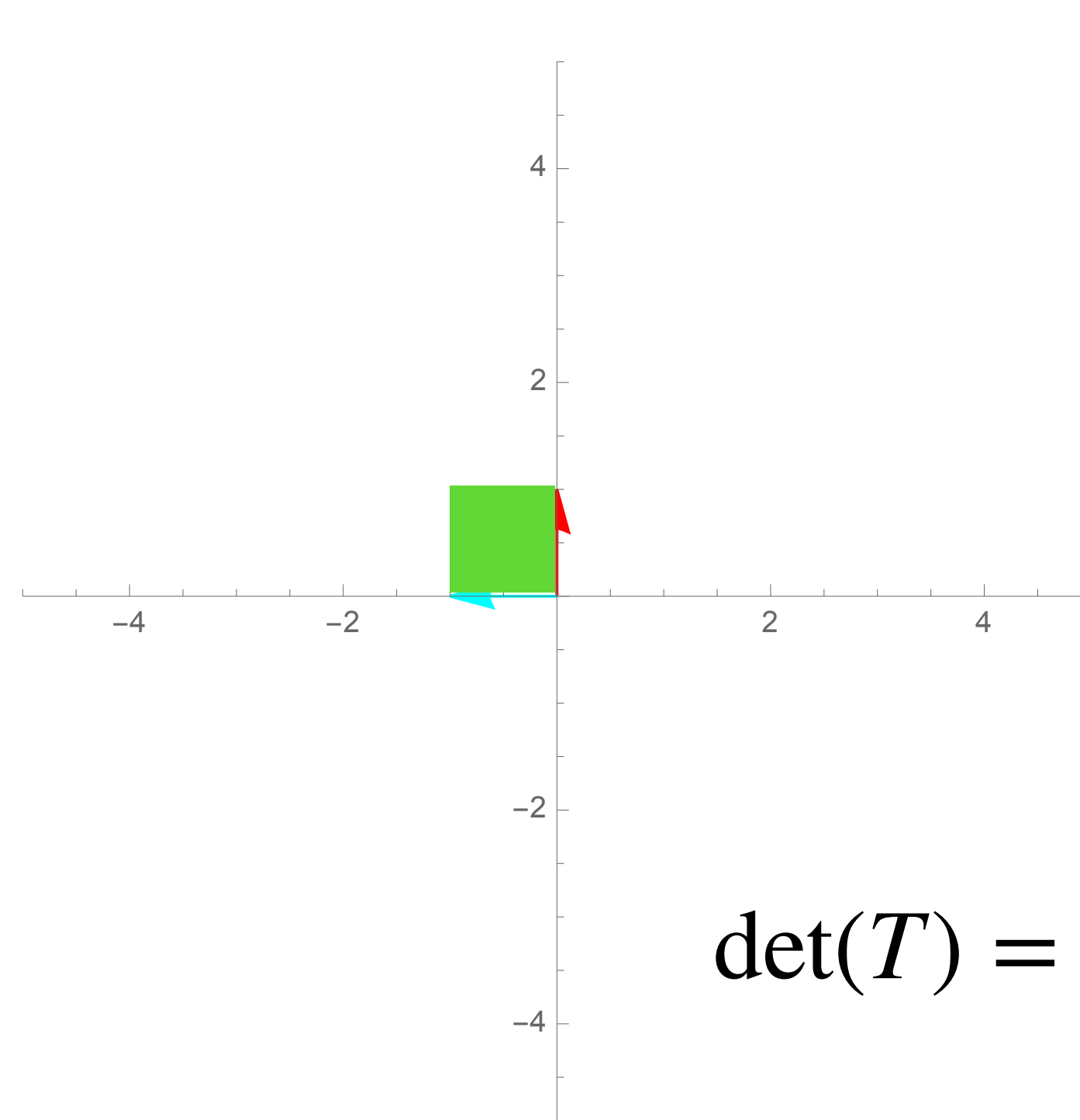
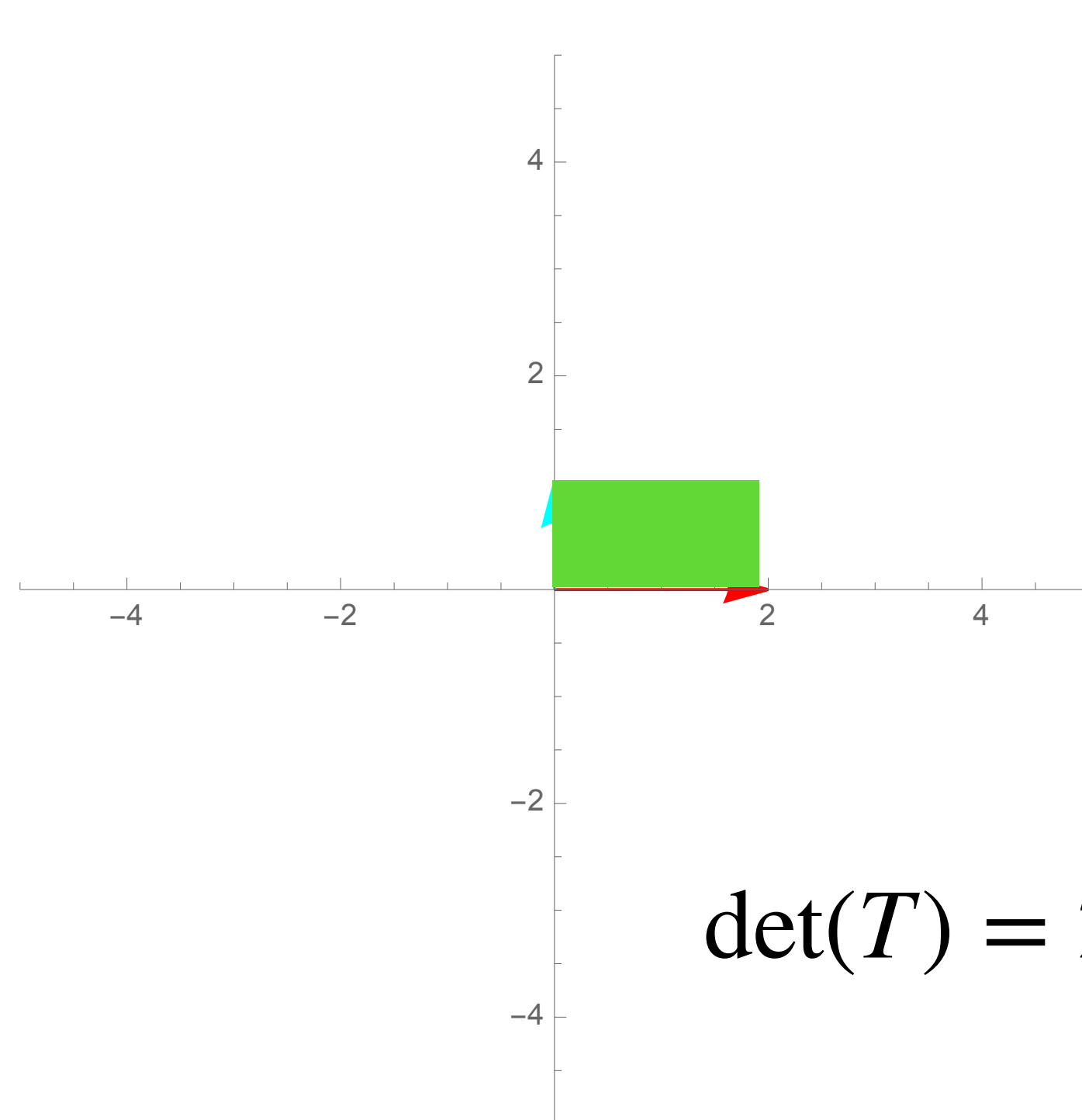


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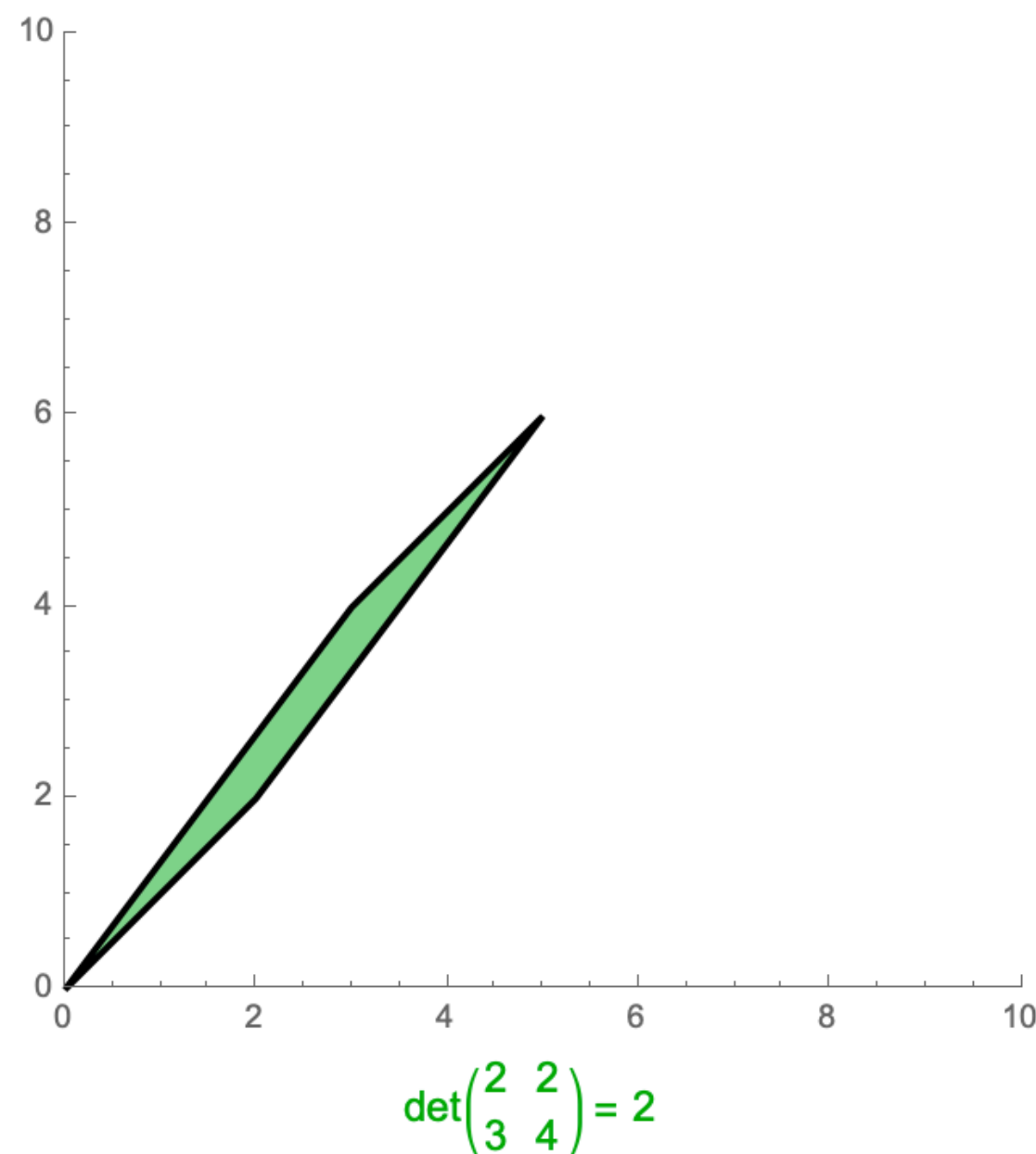
diagonal

How can we calculate $\det(T)$?

What info does $\det(T)$ give us about T ?

What properties do/can/must $\det(T)$ have?

geometry



algebra

$$\det(AB) = \det(A) \det(B)$$

arithmetic

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix}$$

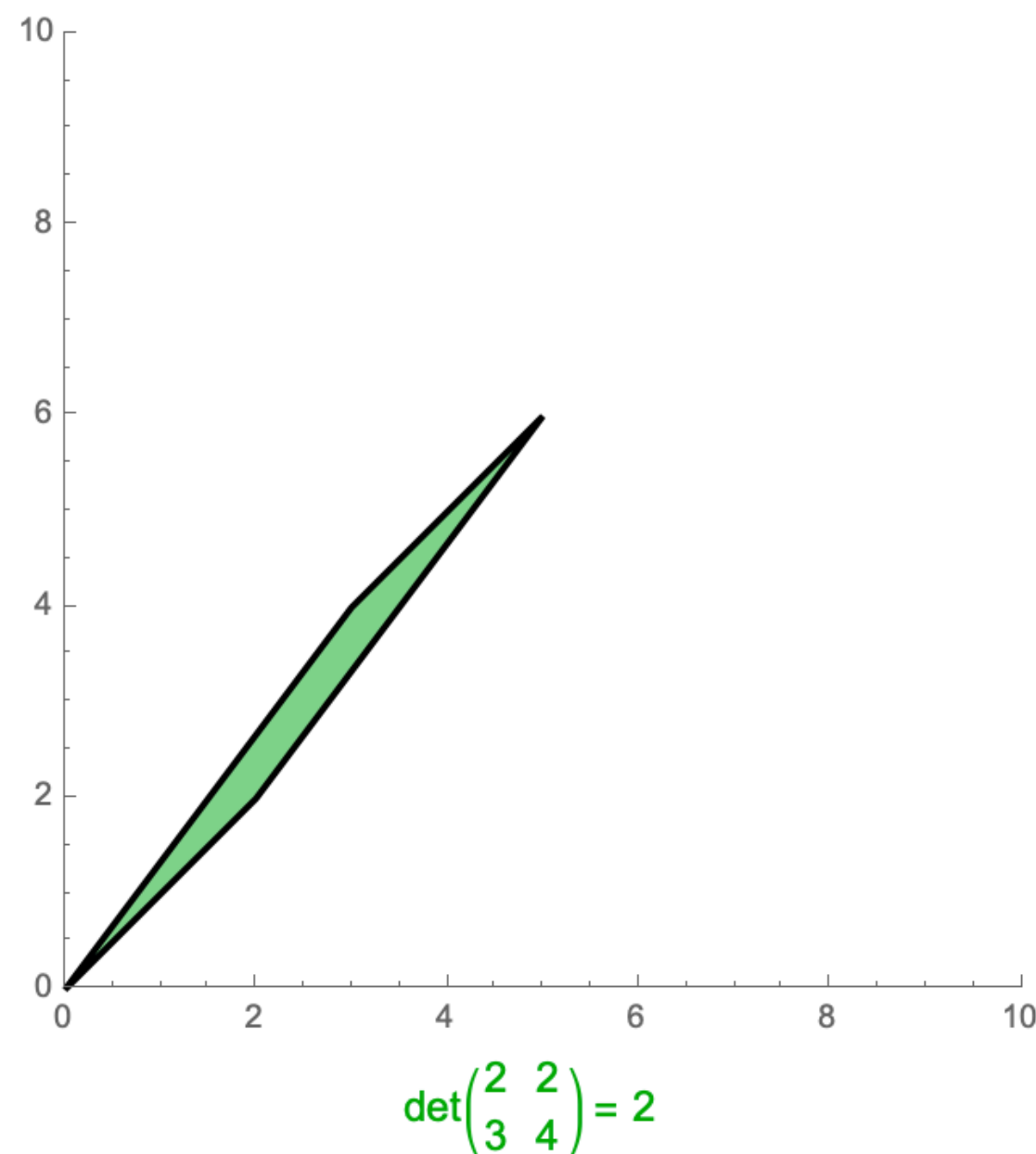
$$= aei + bfg + cdh - afh - bdi - ceg$$

functions

$$M_{n \times n}^{\mathbb{F}} \rightarrow \mathbb{F}$$

multilinear,
antisymmetric, &
normalized

geometry



algebra

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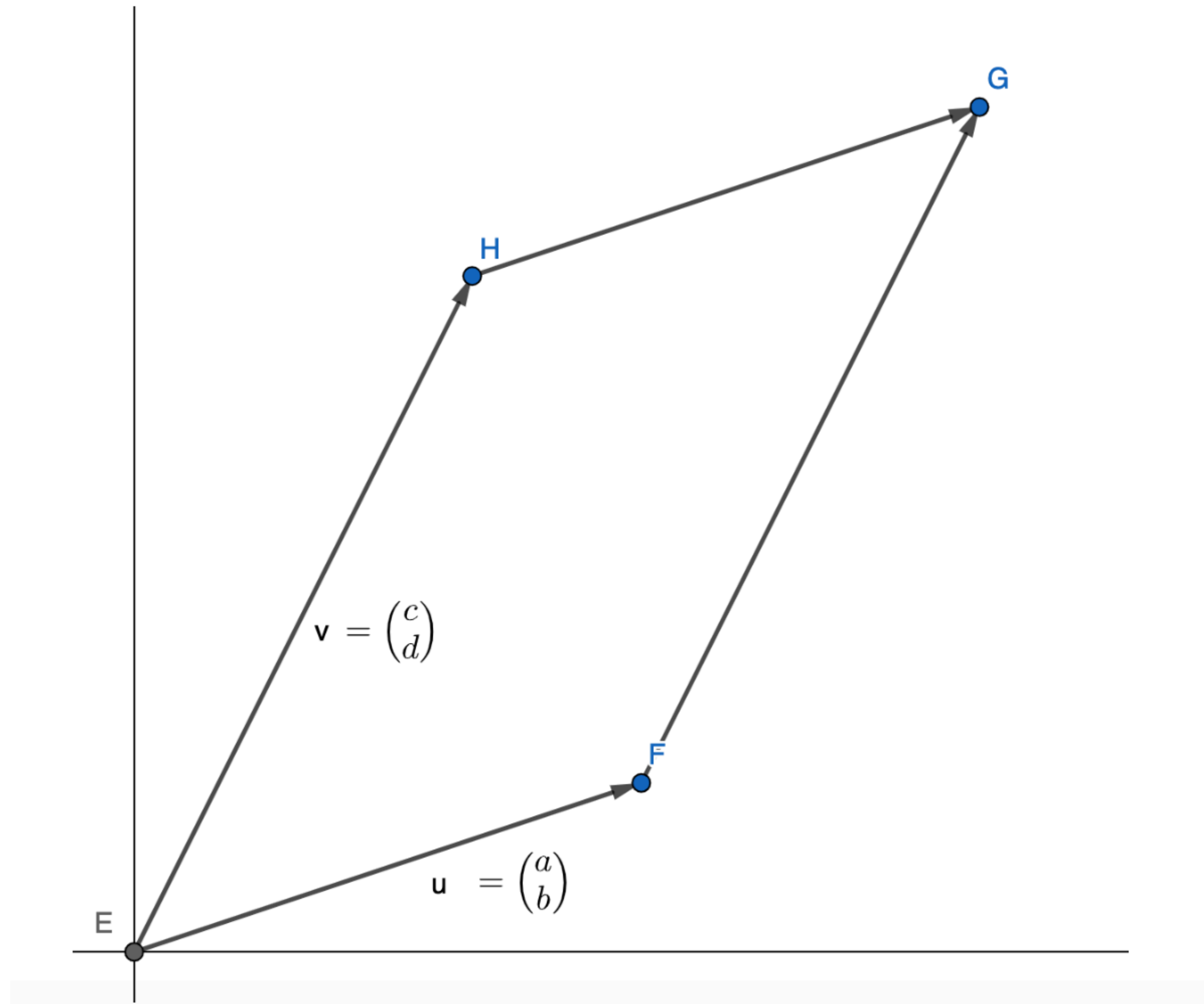
functions

differential forms

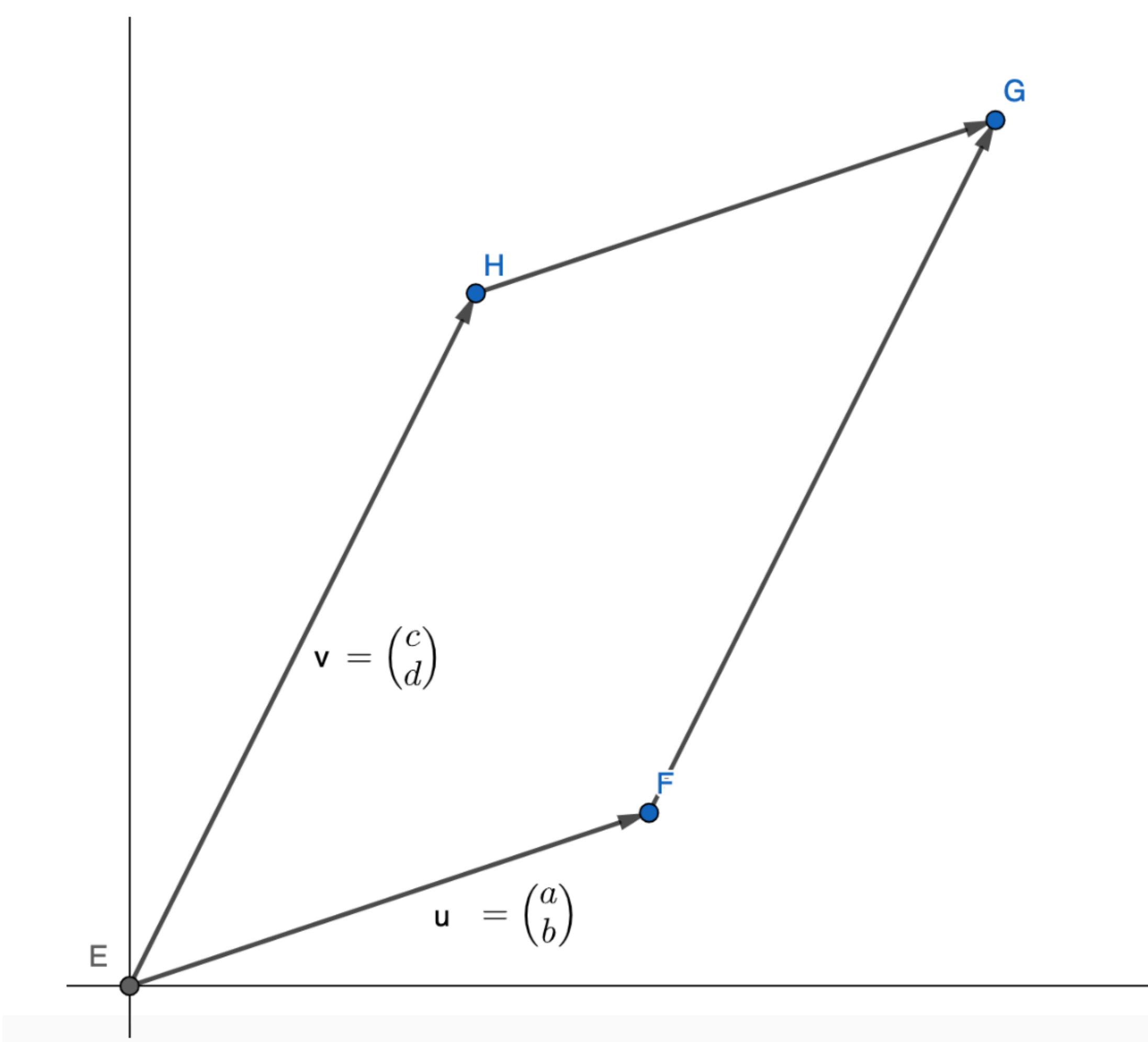
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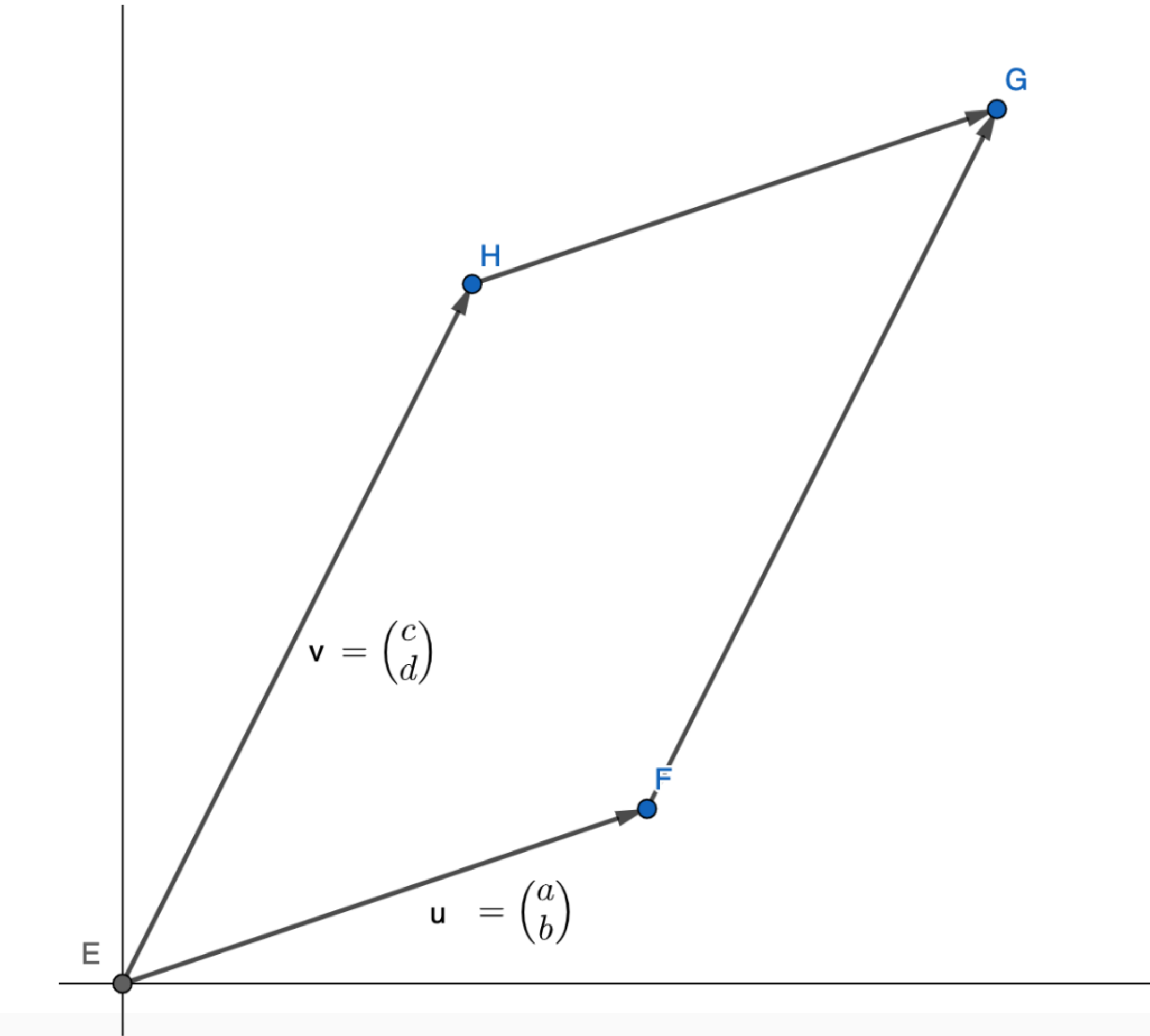
Calculate the area of parallelogram $EFGH$.



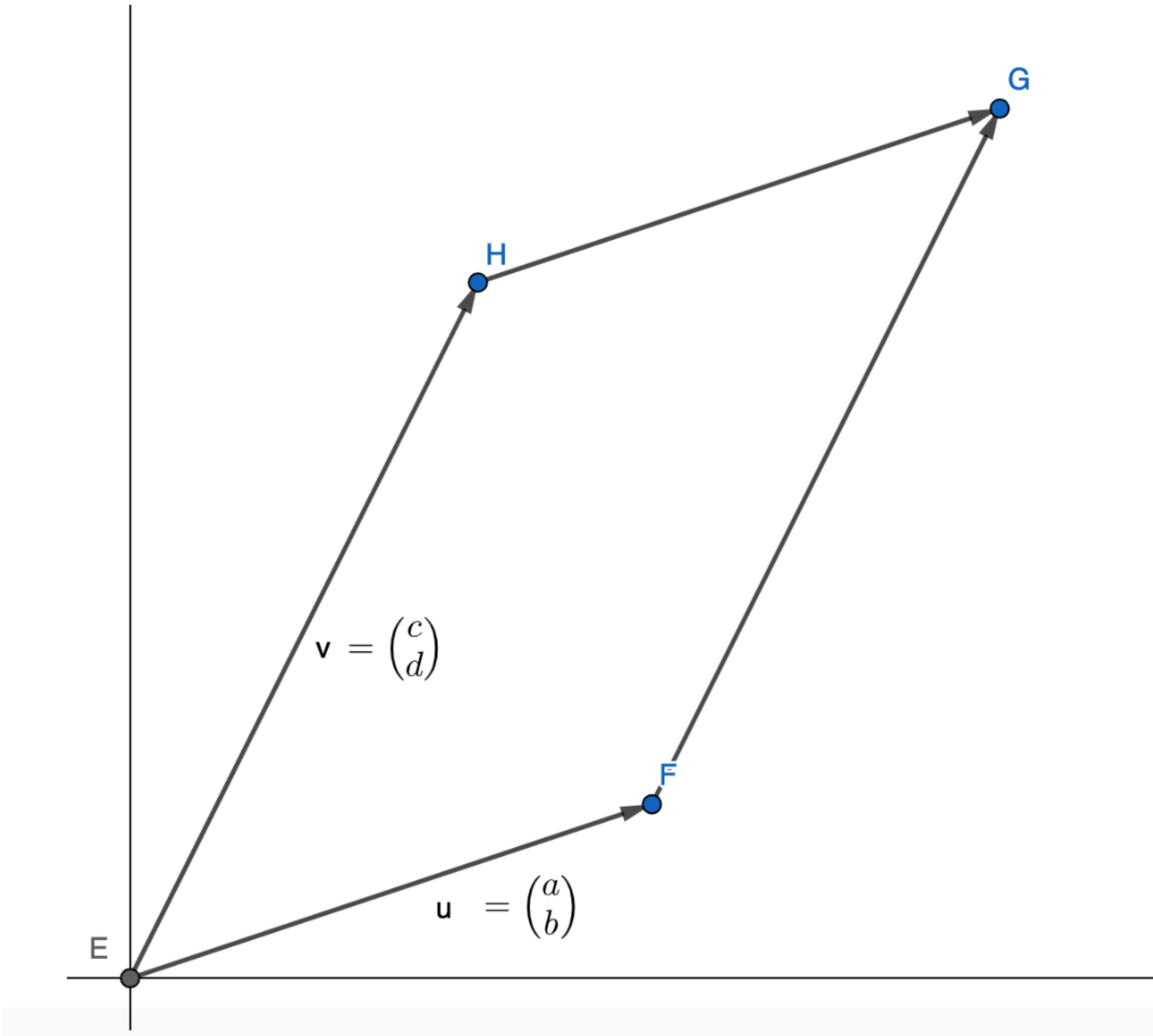
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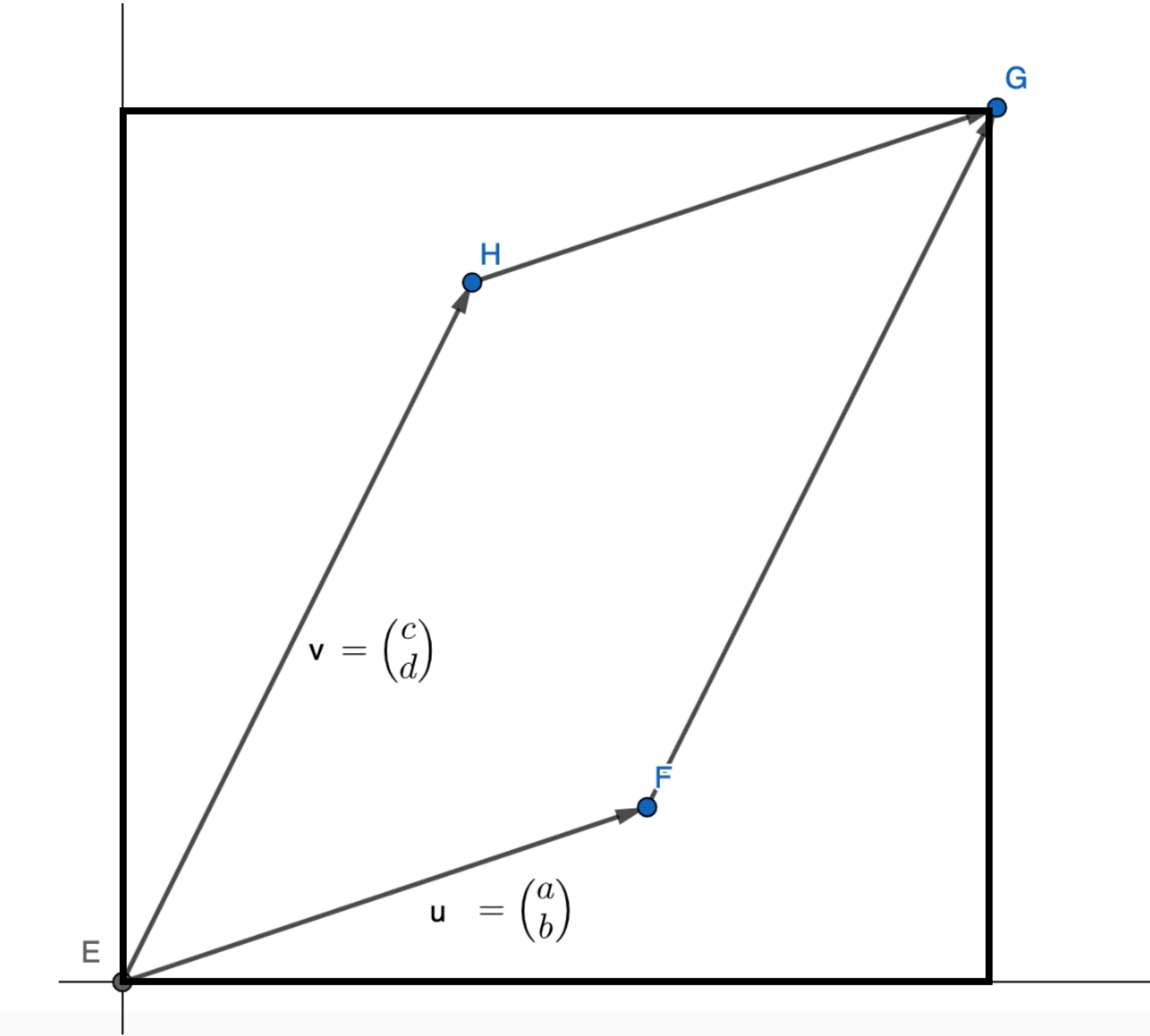
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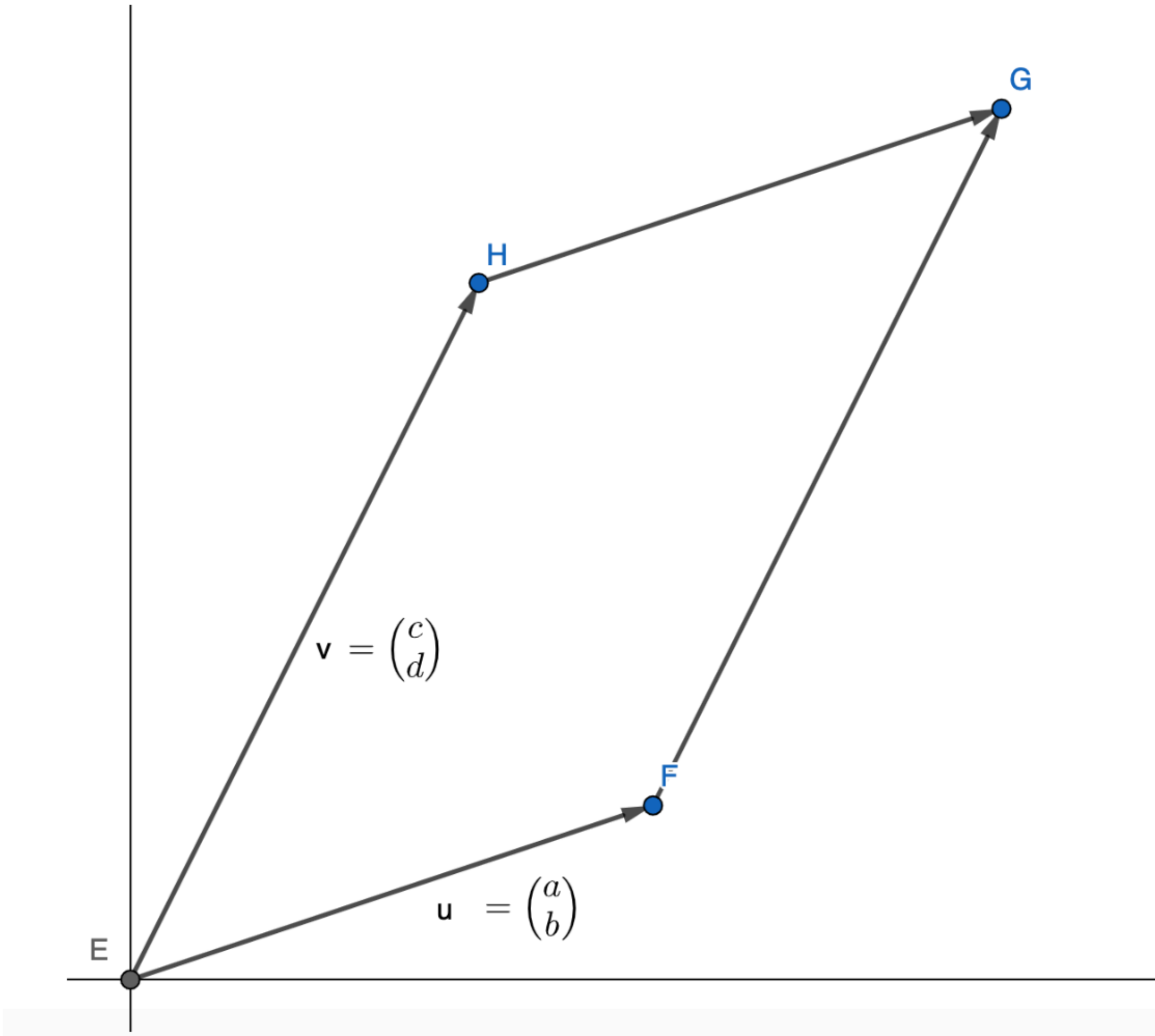
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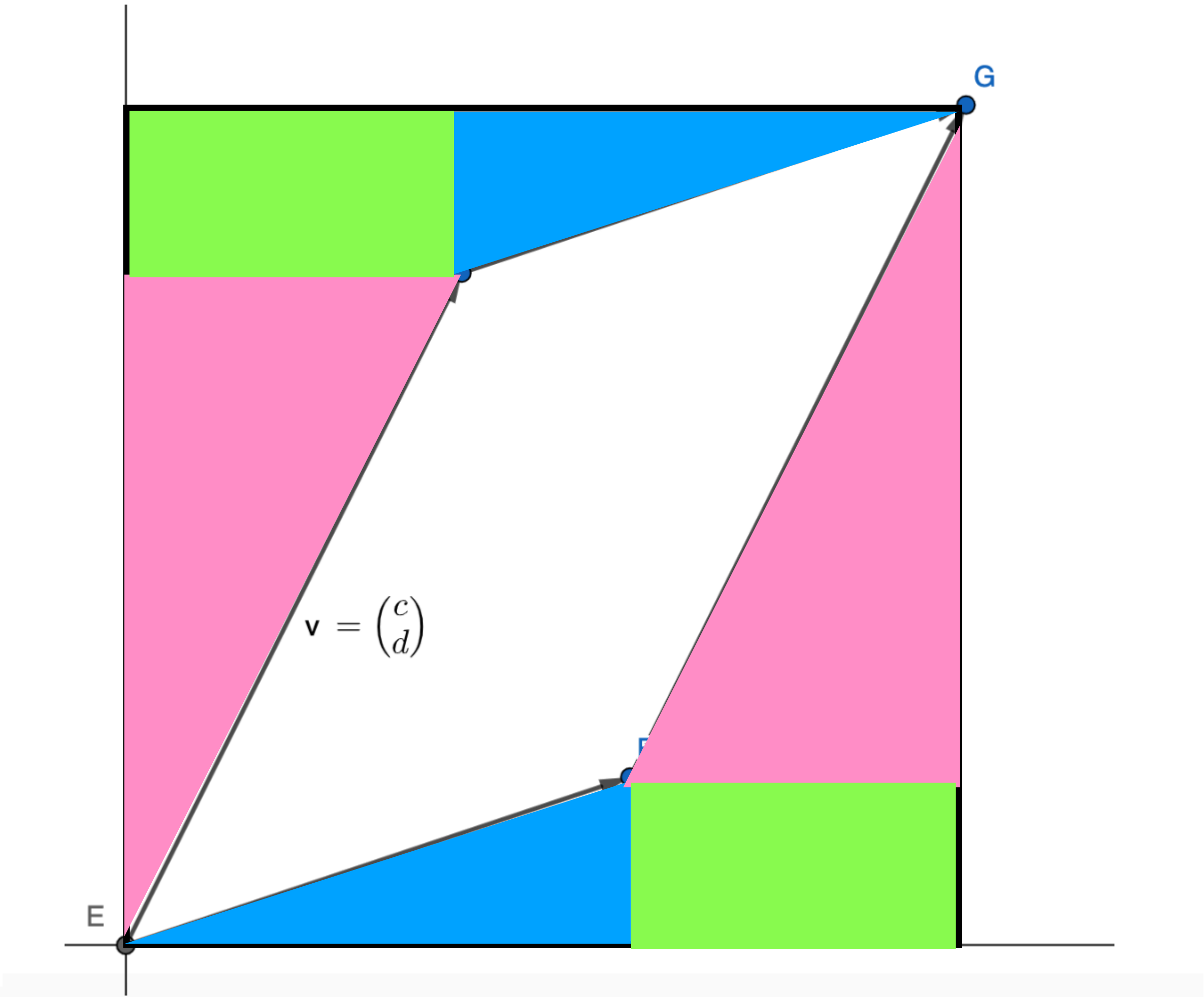
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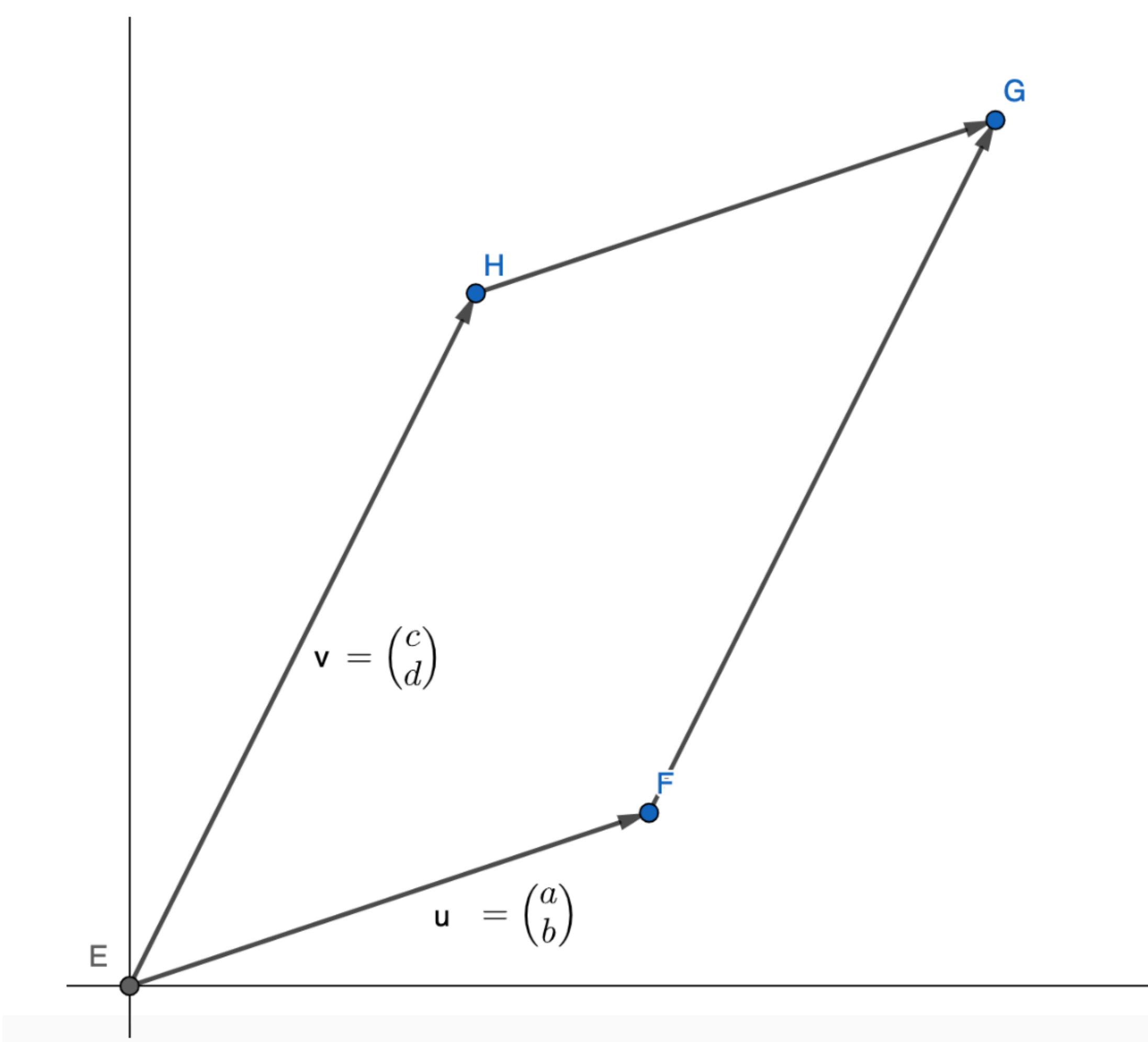
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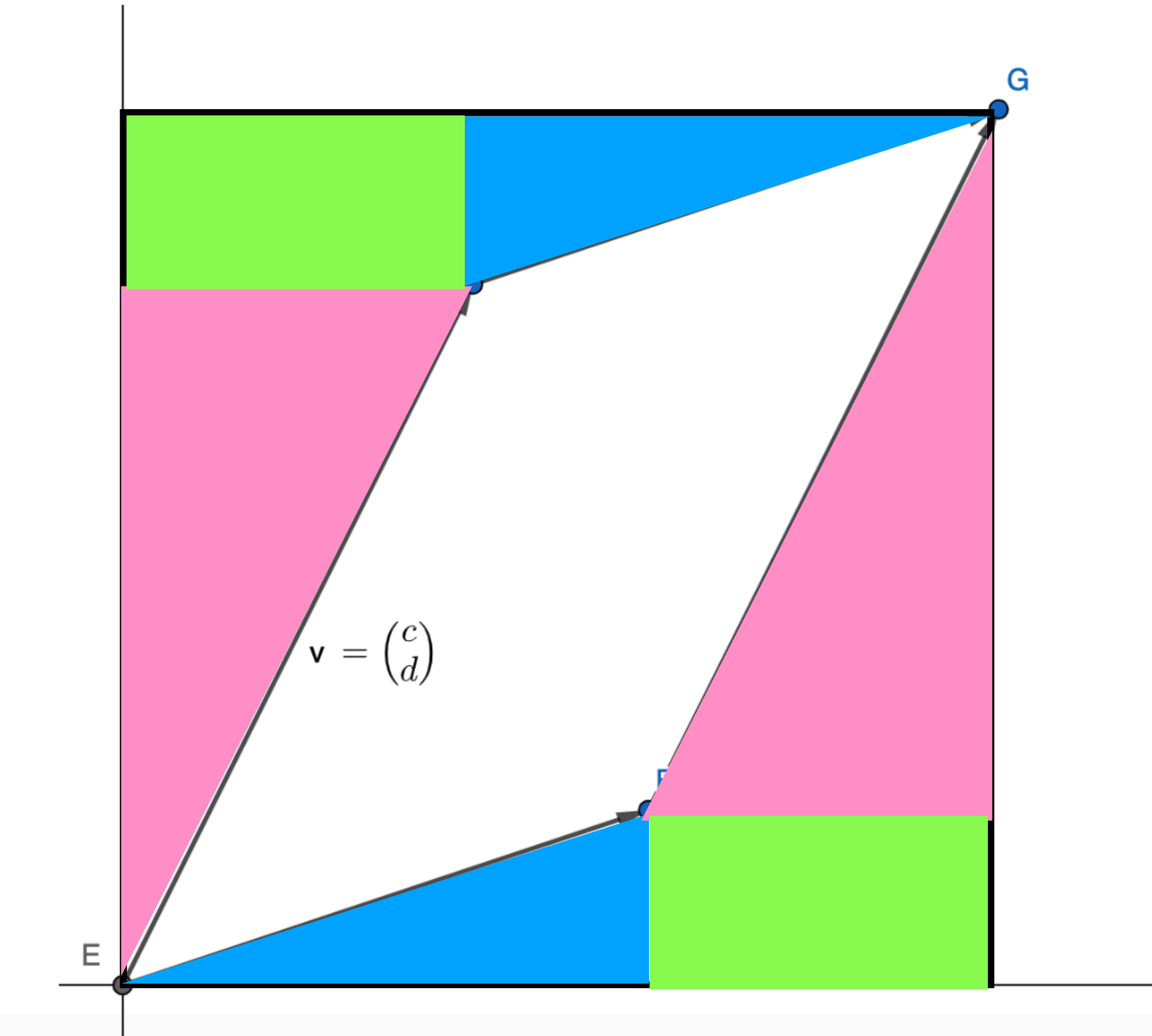
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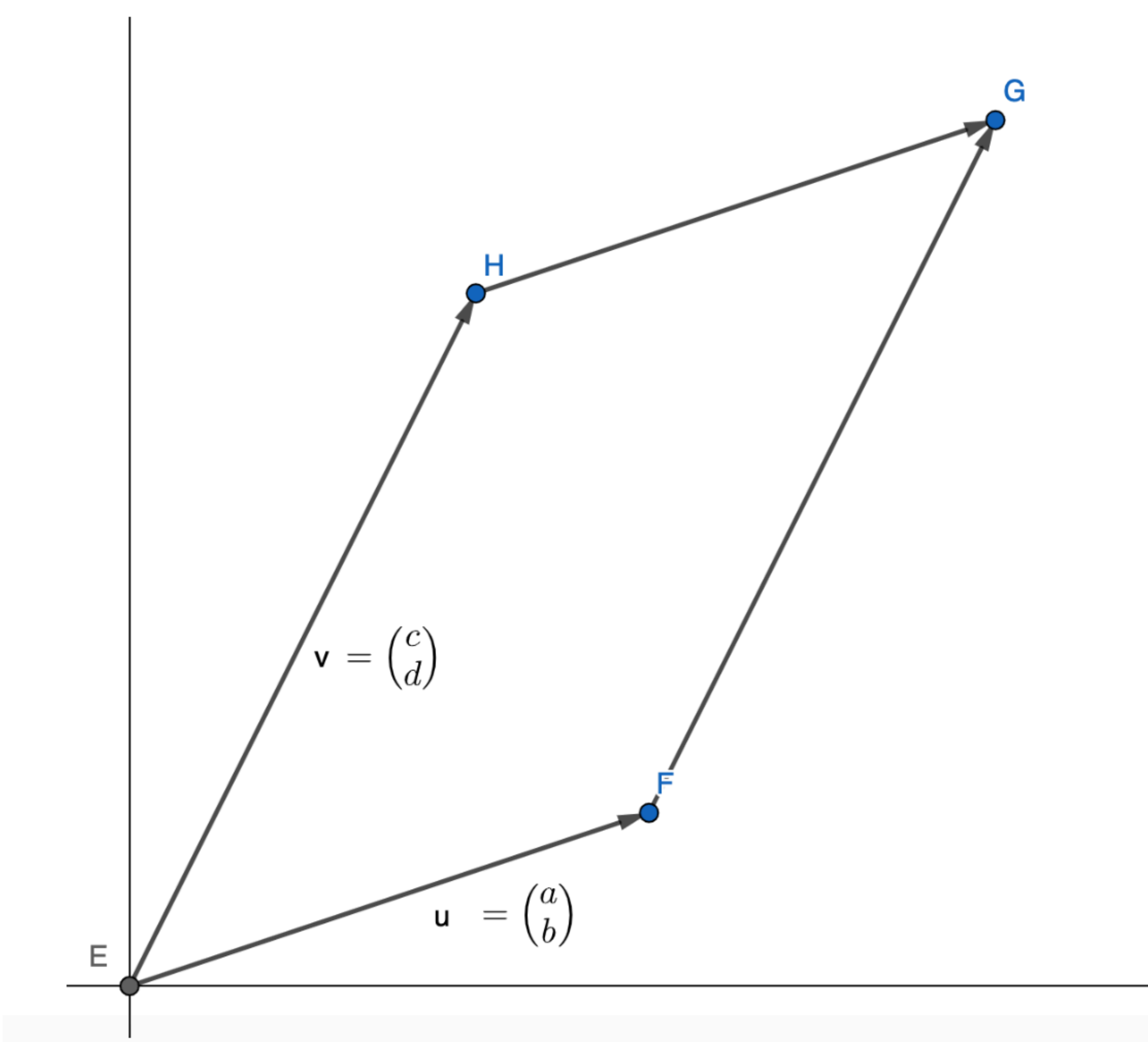


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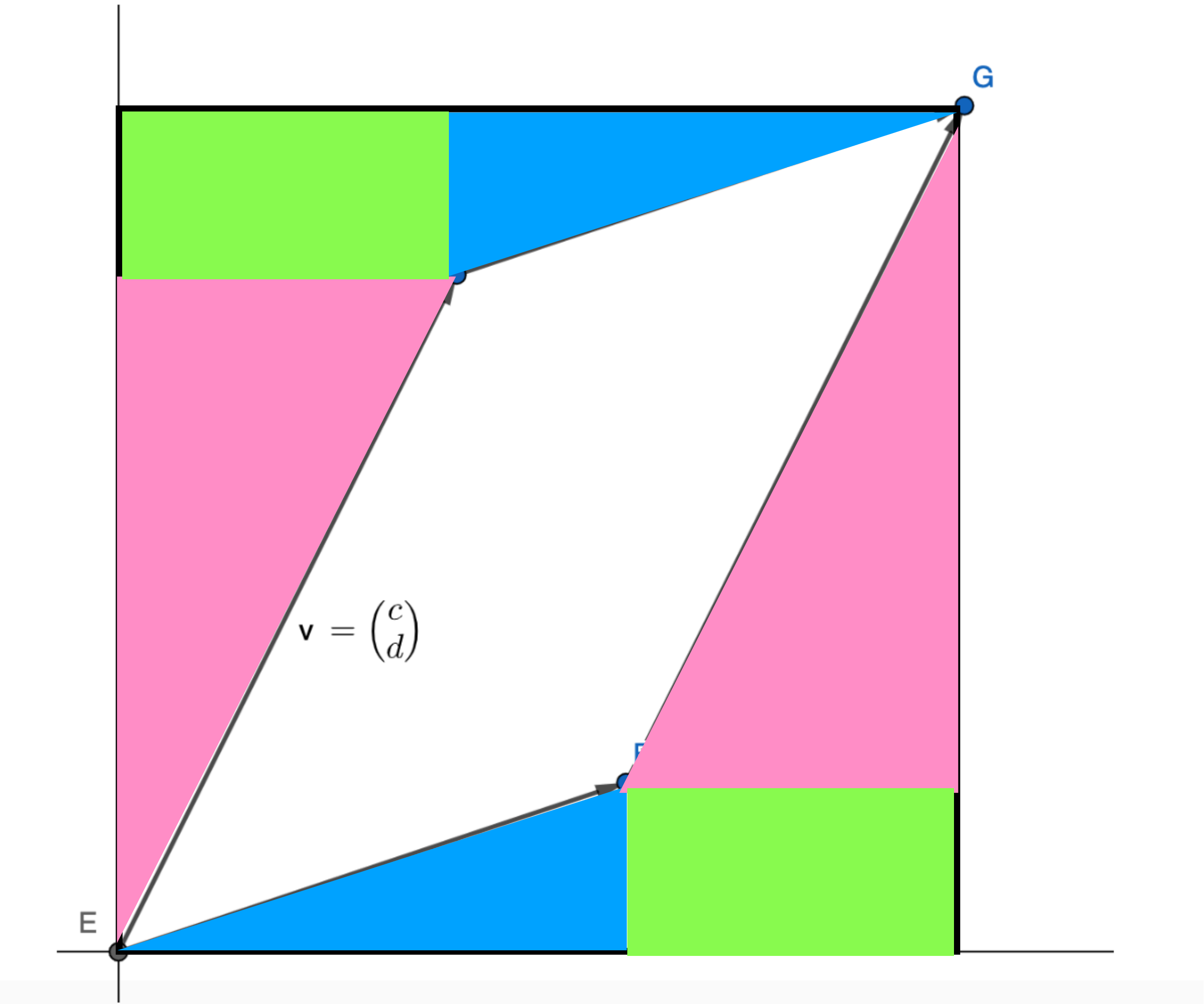


$$(a+c)(b+d) - ab - cd - bc - bc =$$

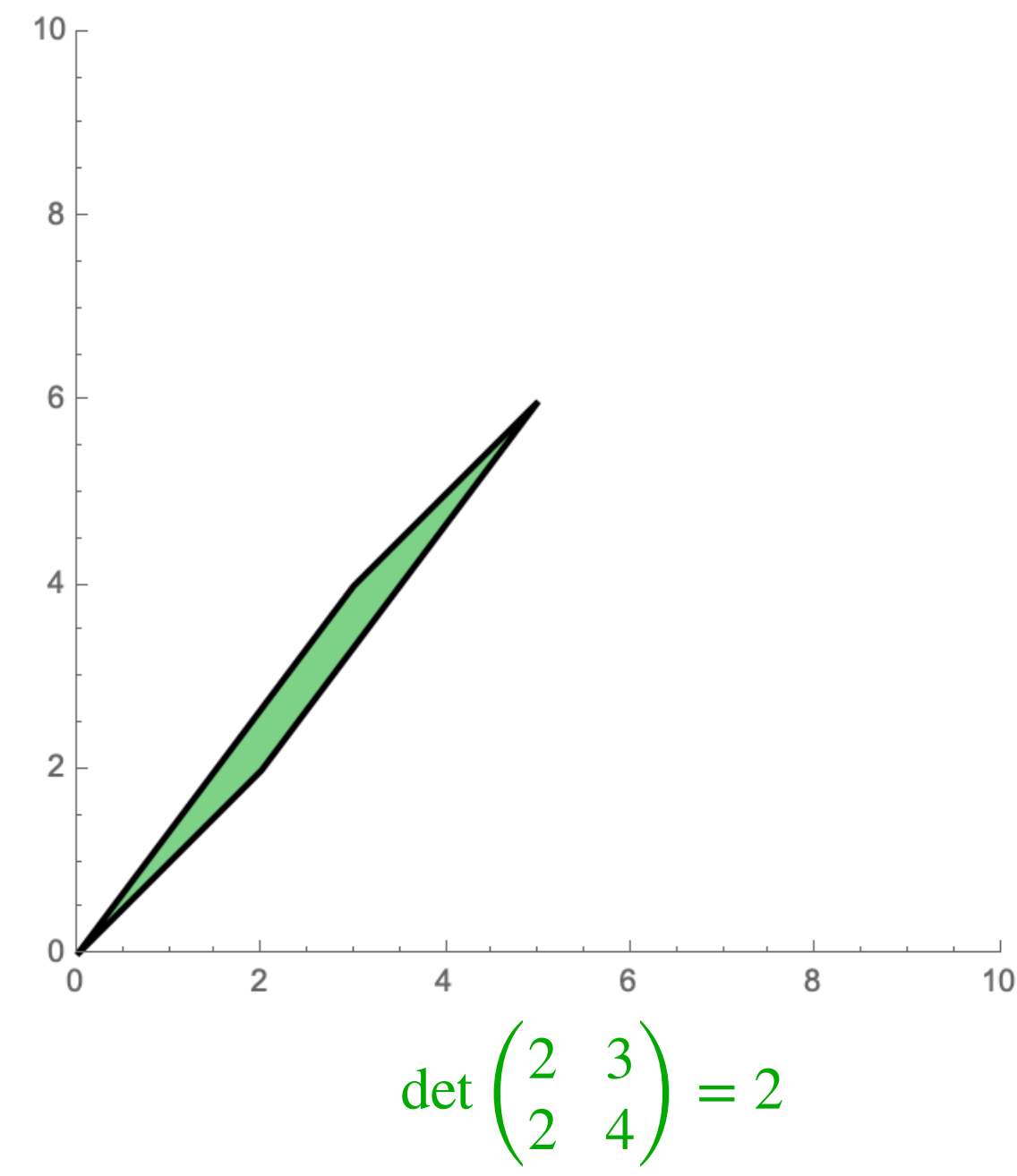
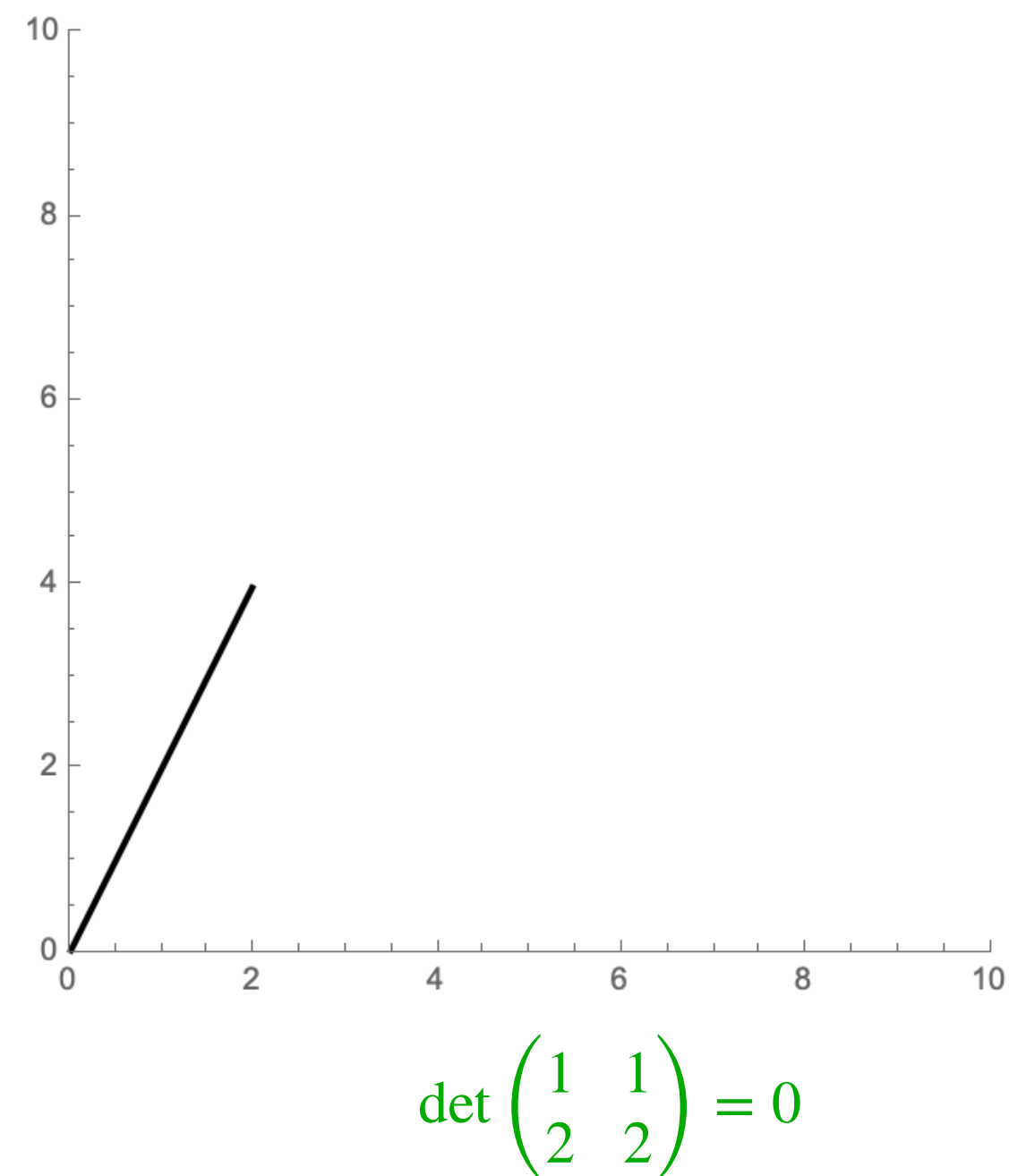
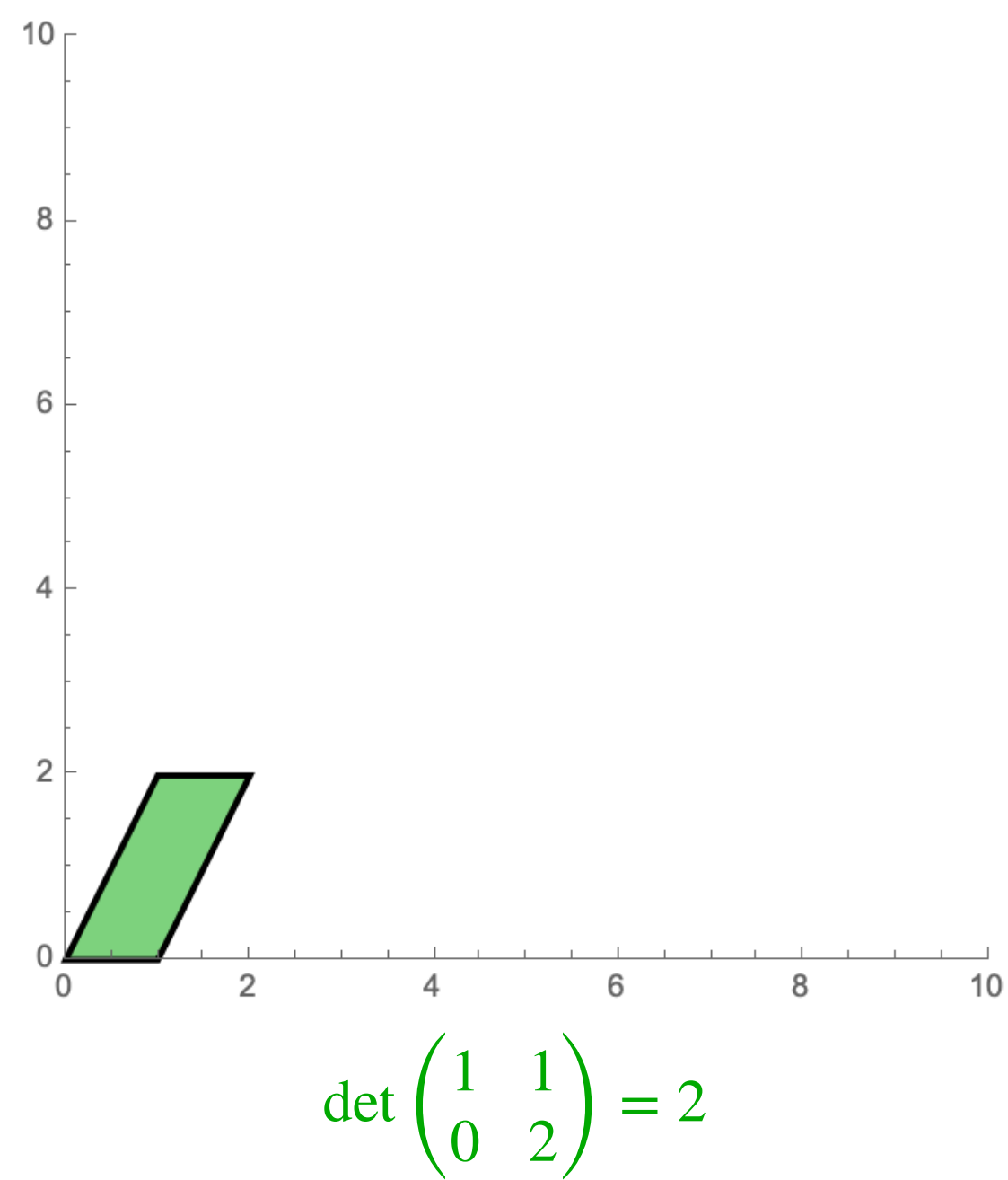
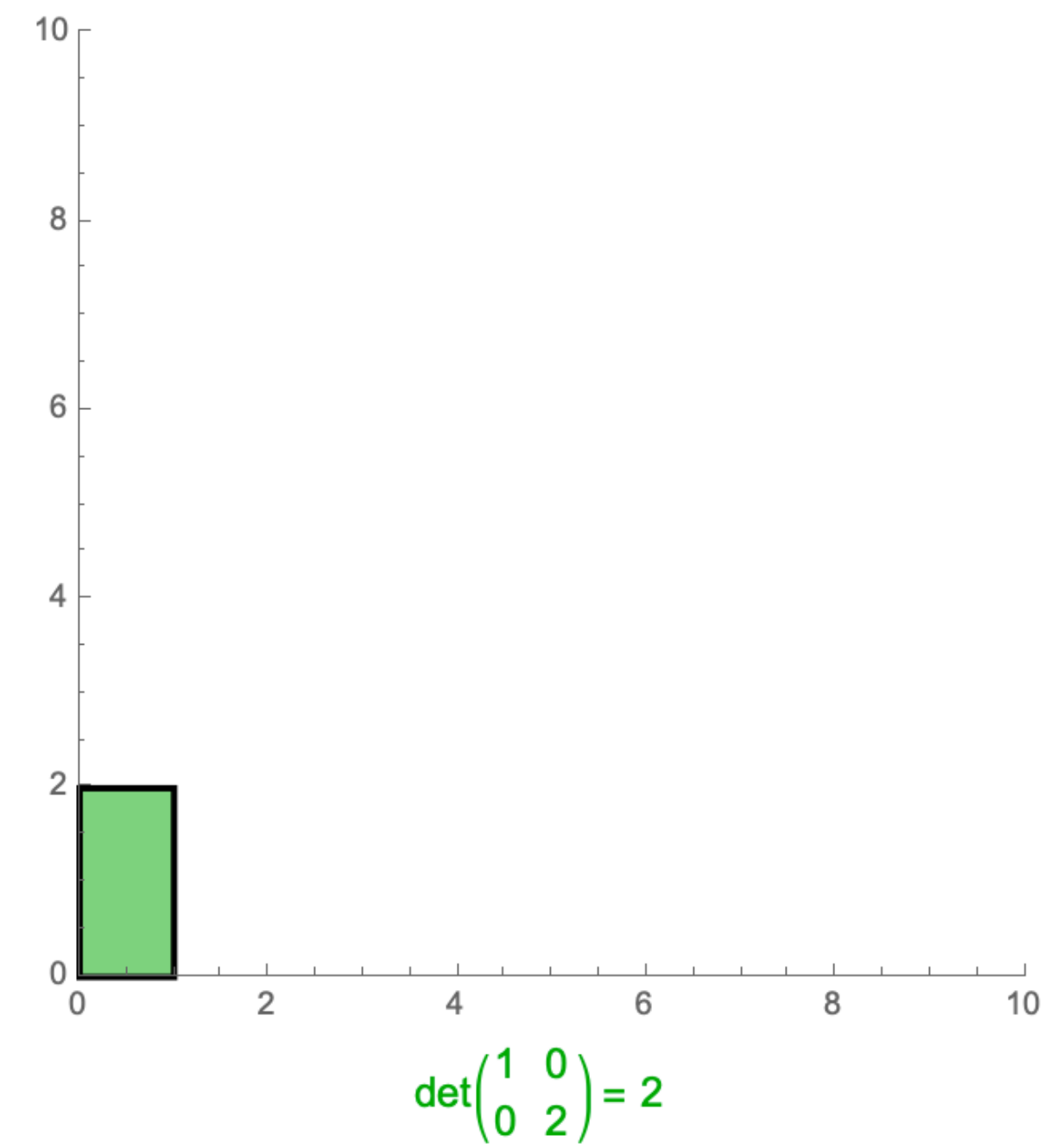
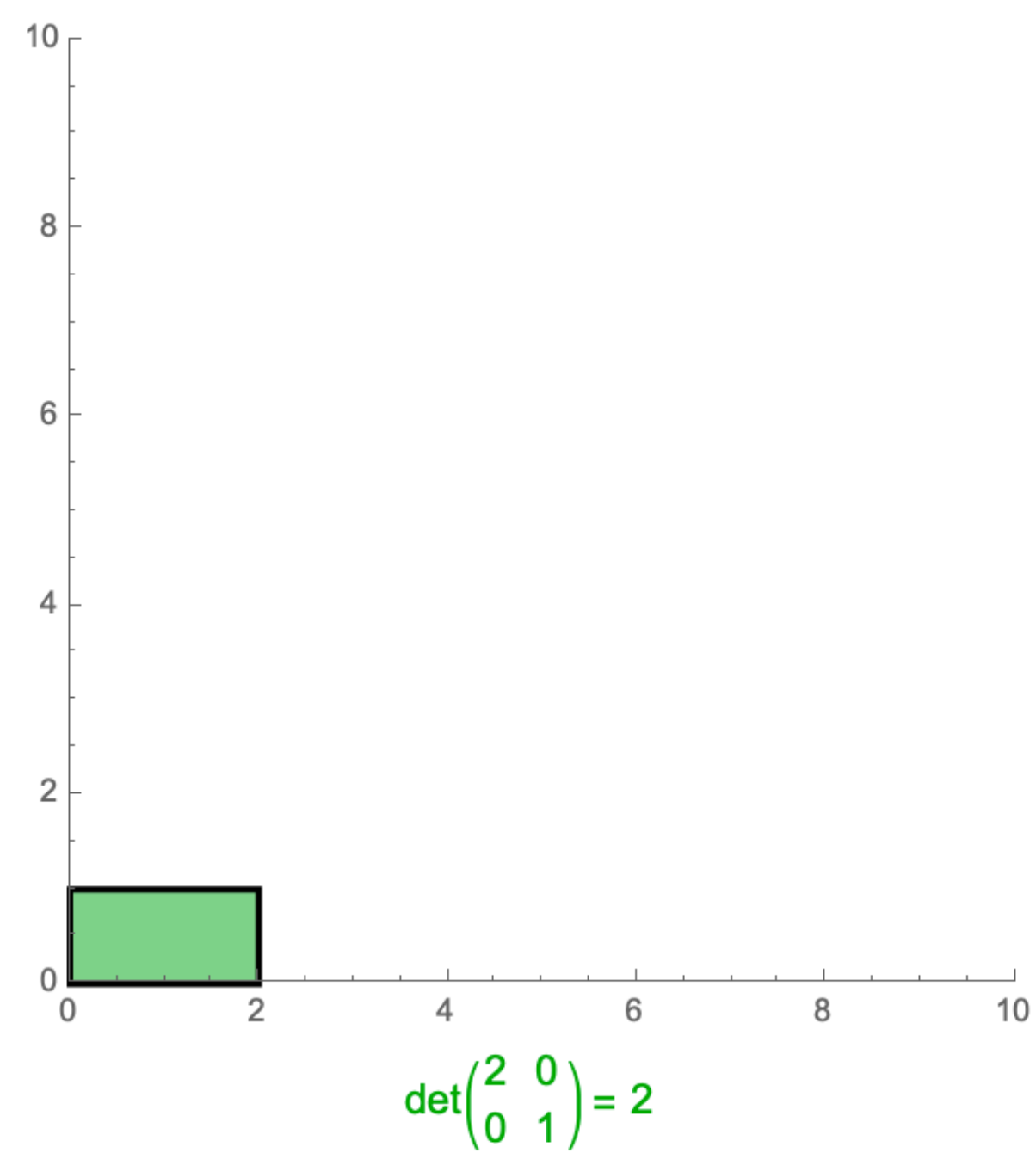
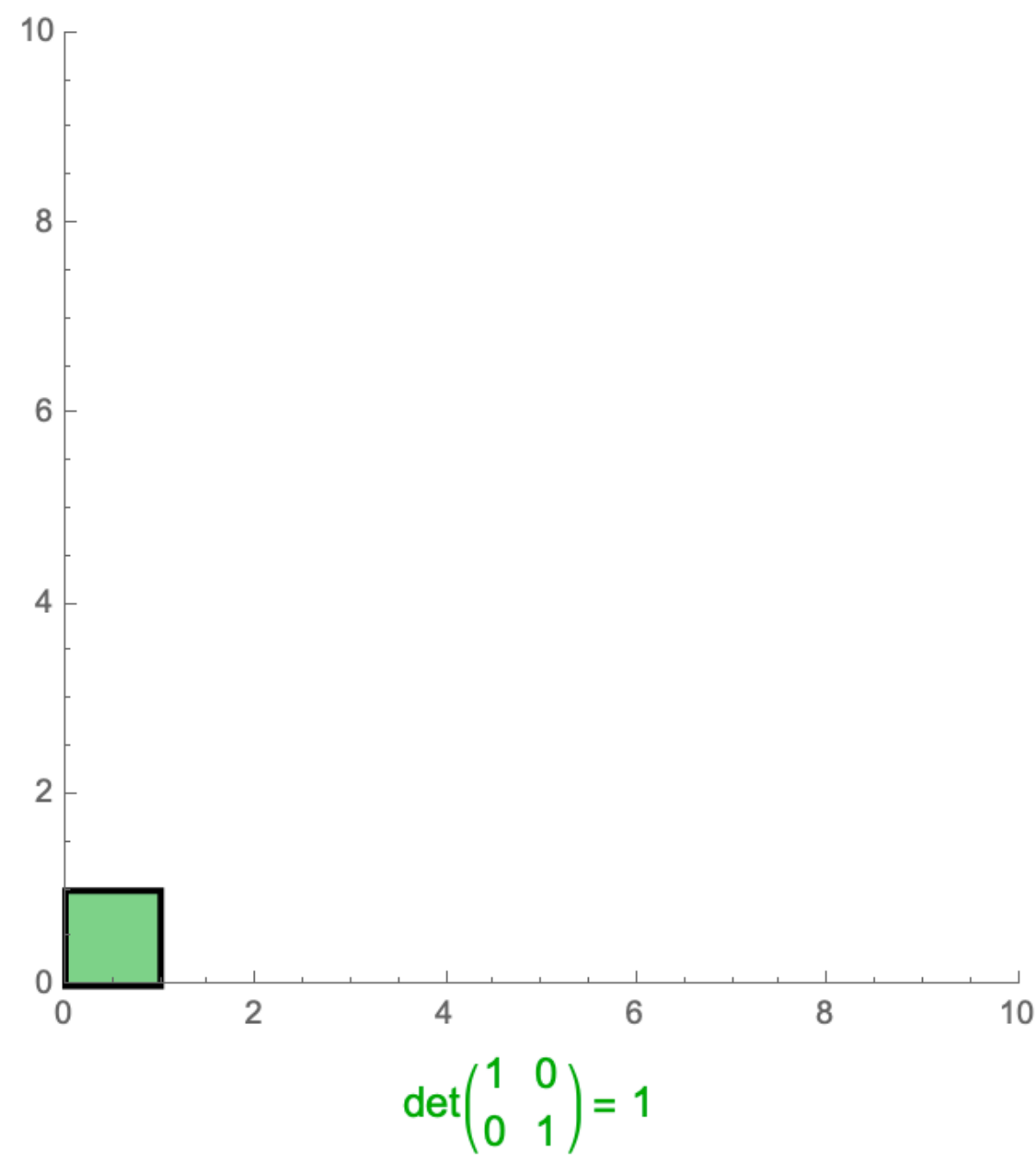
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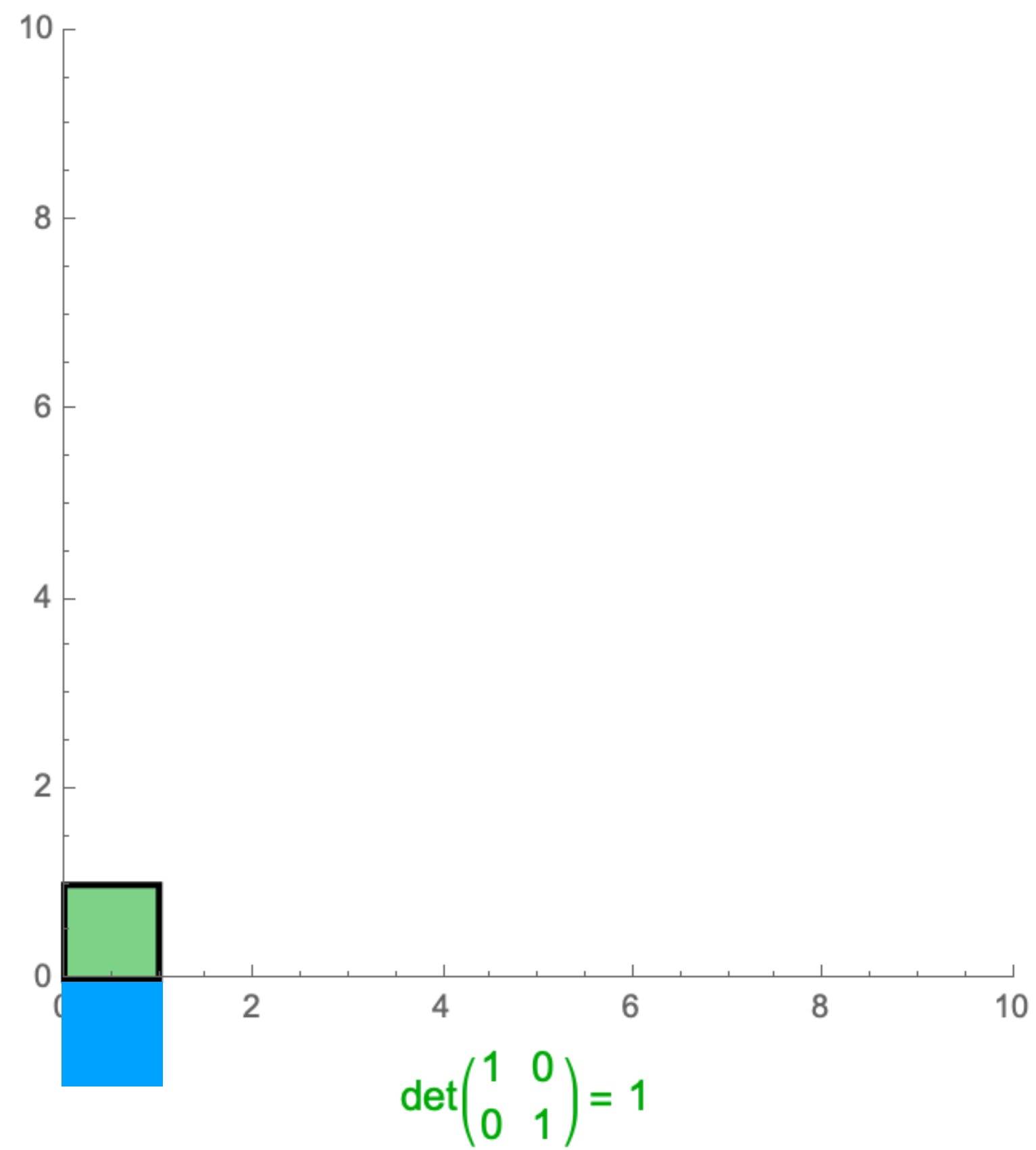
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$$(a+c)(b+d) - ab - cd - bc - bc = ad-bc$$

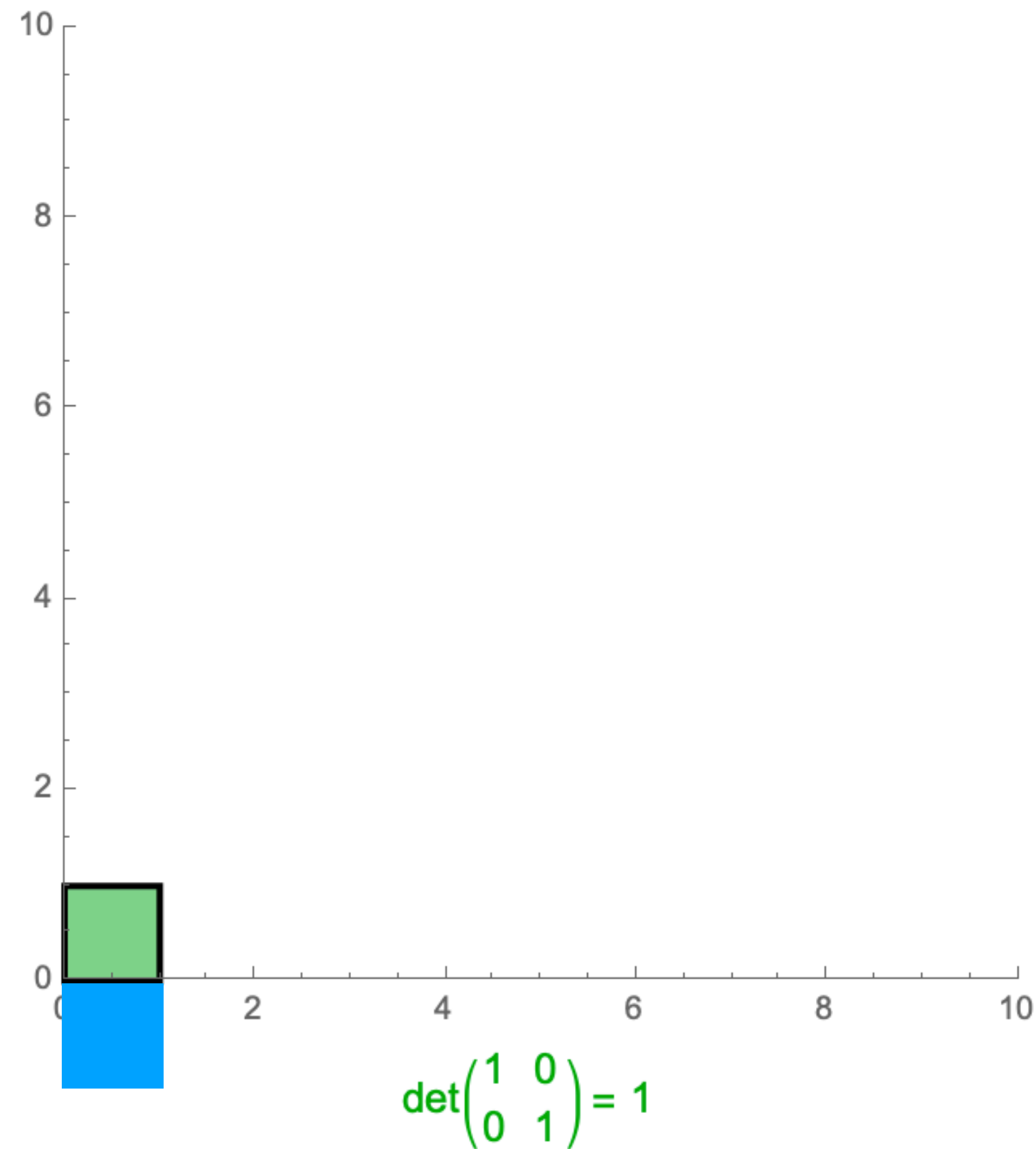


orientation



$$\det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1$$

orientation



$$\det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1$$

reflections have determinant -1

orientation preserving isometries have determinant 1

orientation *reversing* isometries have determinant -1

projections have determinant 0

Make up an arbitrary function f in three real variables x , y , and z .

For your function, calculate $f(1, 2, 3)$, $f(2, 2, 3)$ and $f(2, 1, 3)$.

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multilinear? (anti)-symmetric? normalized?

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multilinear: scaling an entry by k scales the output by k

(anti)symmetric: swapping two entries preserves (negates) the output

normalized: the output for one particular input is 1

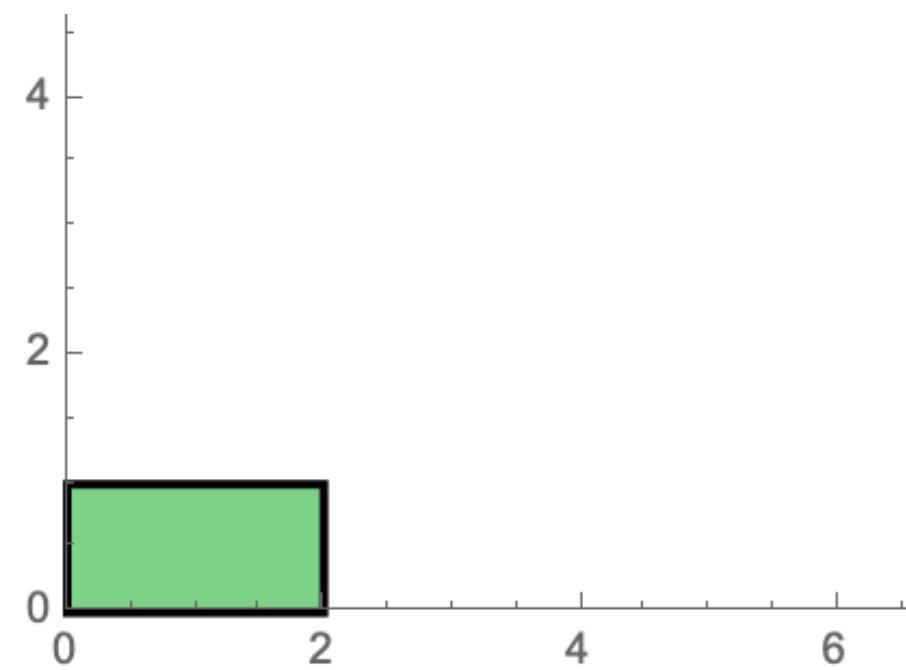
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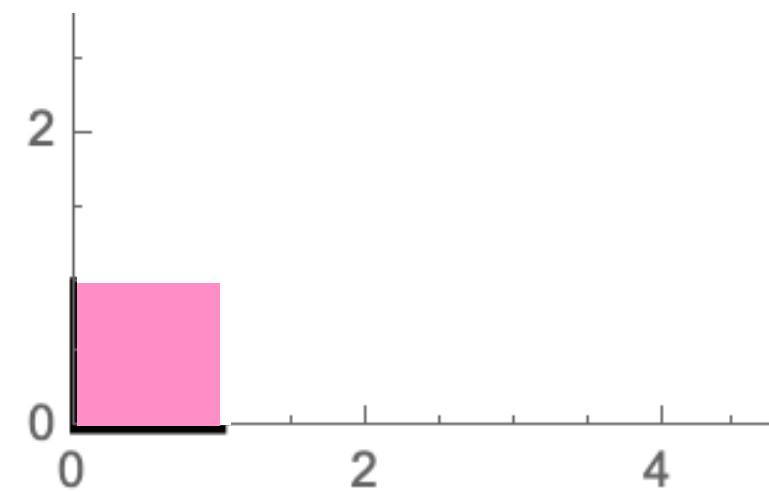
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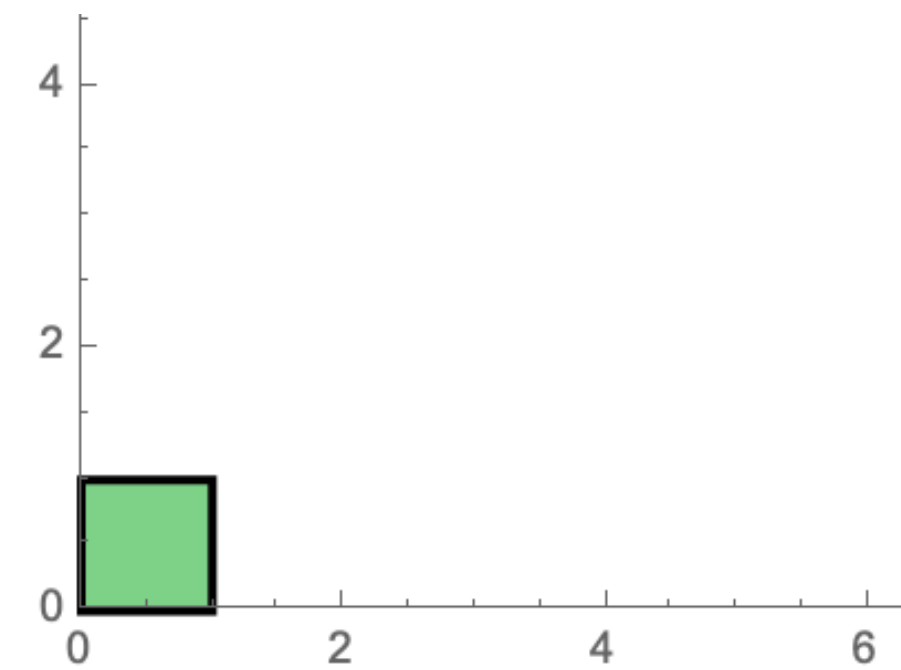
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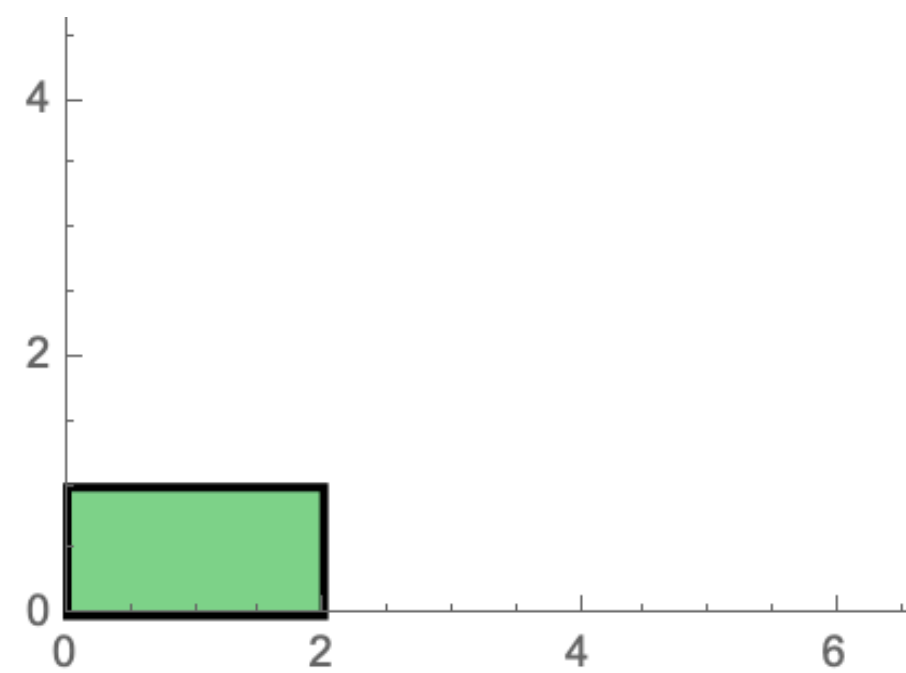
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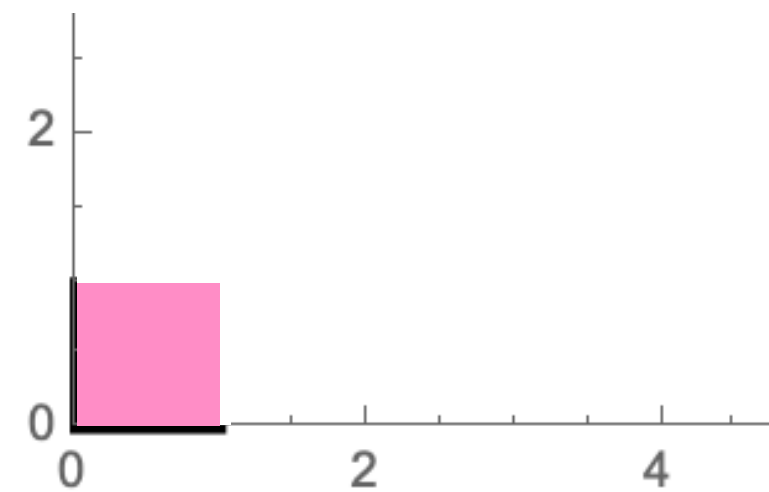
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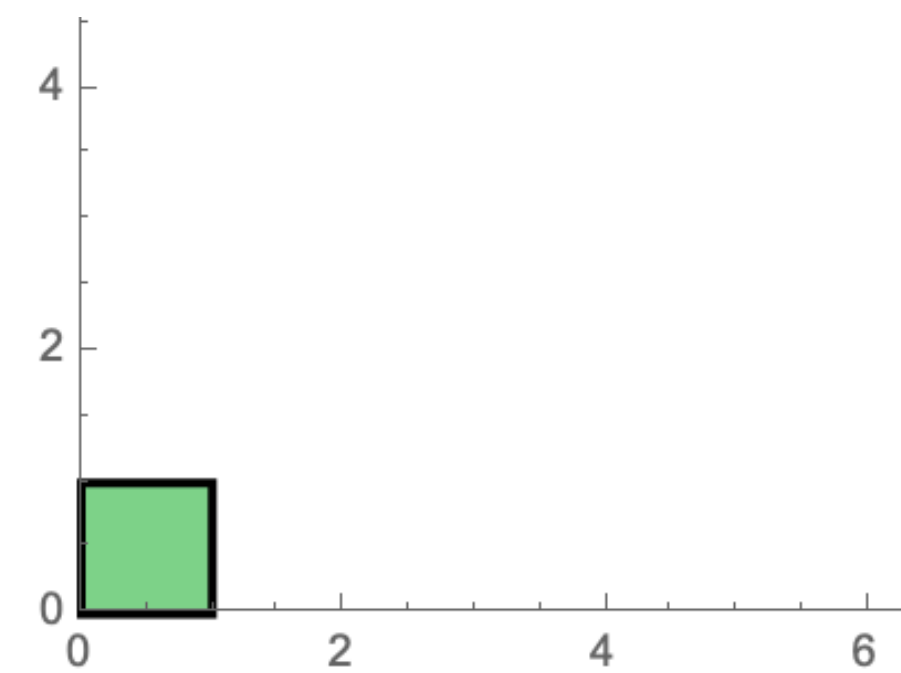
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$$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$$

There exists a unique map $M_{n \times n}^{\mathbb{F}} \rightarrow \mathbb{F}$
that is multilinear, antisymmetric and normalized,
which is the determinant.

$$\det 3A \stackrel{?}{=} 3 \det A$$

$$\det 3A \neq 3 \det A$$

$$3 \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 3$$

$$\det \left(3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \det \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = 9$$

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Today: use defining properties to calculate determinants

Three methods for computing matrix inverses (when they exist):

- use geometric intuition
- use row reduction
- use a formula

Four methods for computing the determinant of a matrix:

- use geometric intuition
- use row reduction
- use a formula
- notice a special form
- use cofactor expansion

5 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix
- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
 $A=LU$ and $A=PLU$ and $A=PLDU$ factorizations

6 things (so far!) that row reduction is great for:

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Column operations on a matrix affect its determinant in predictable ways.

scaling a column by k scales the determinant by k

adding a multiple of a column to a given column preserves the determinant

swapping two columns negates the determinant

the identity matrix has determinant 1

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$$\det A = \det A^T$$

So corresponding row operations
have the same effects.

Find the determinant of $\begin{pmatrix} 5 & 4 \\ 3 & 2 \end{pmatrix}$

Find the determinant of $\begin{pmatrix} 5 & 4 \\ 3 & 2 \end{pmatrix}$

Find the determinant of $\begin{pmatrix} 3 & 4 \\ 0 & 1 \end{pmatrix}$

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